

FIG. 1

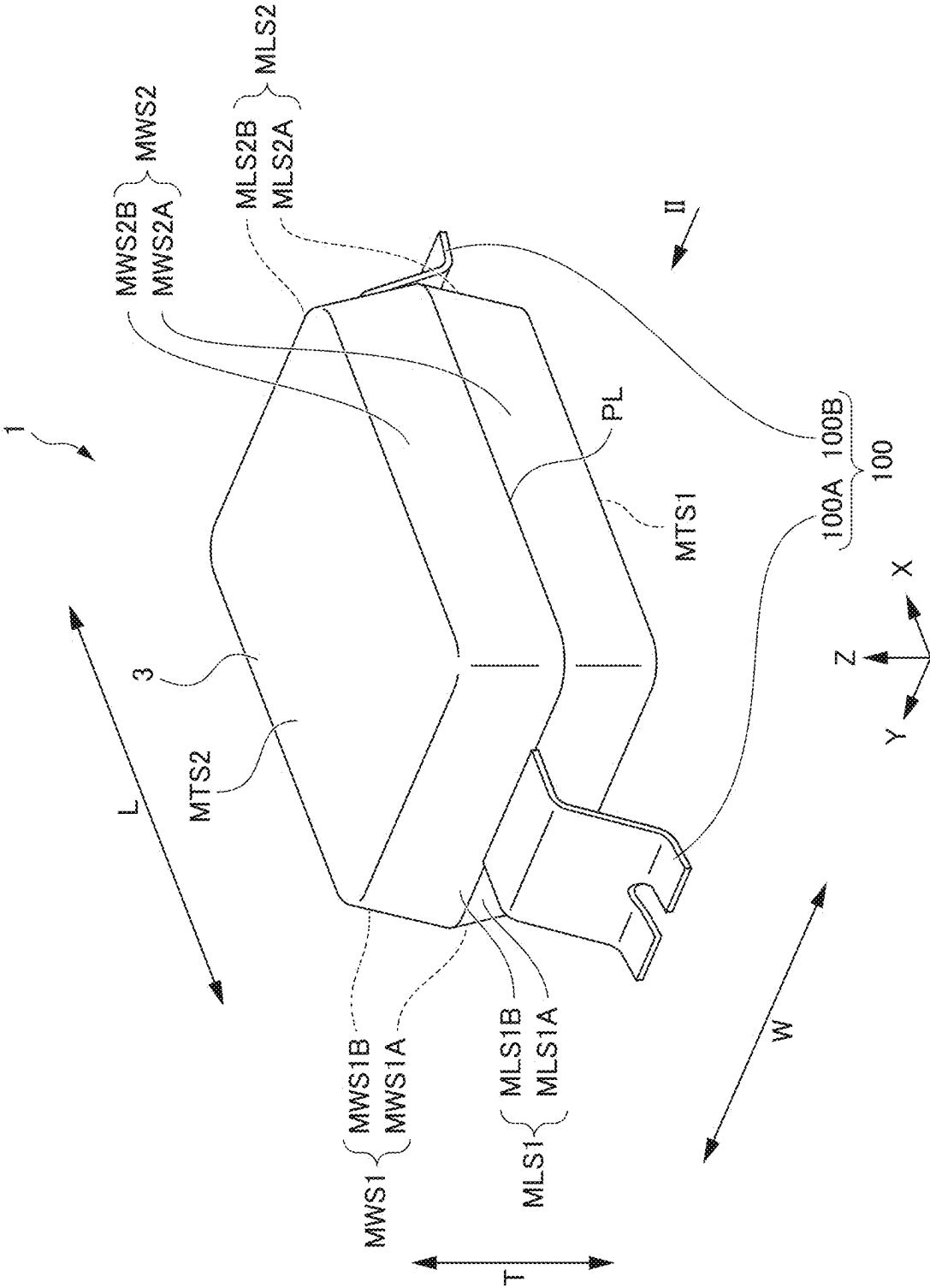


FIG. 2

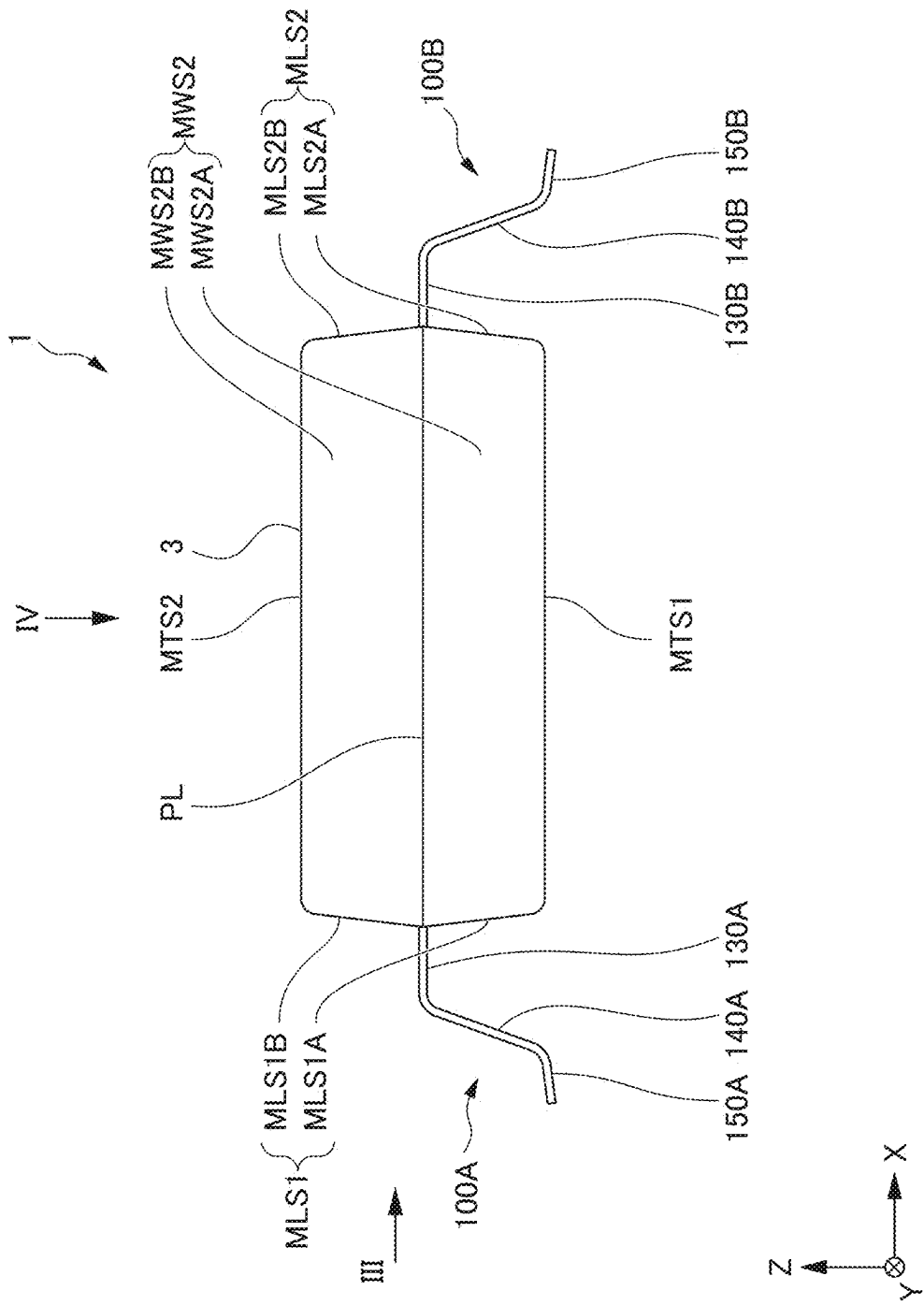


FIG. 3

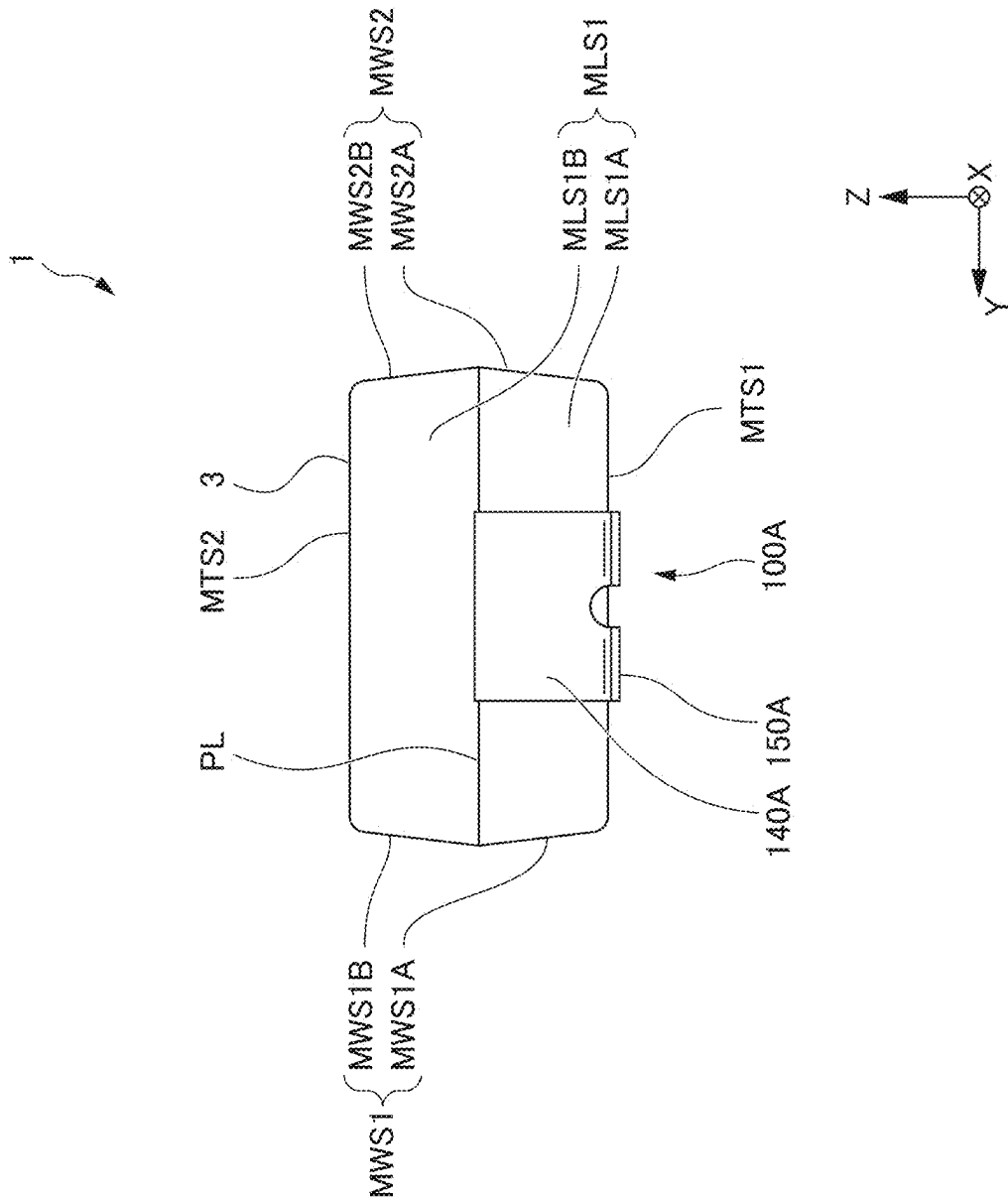
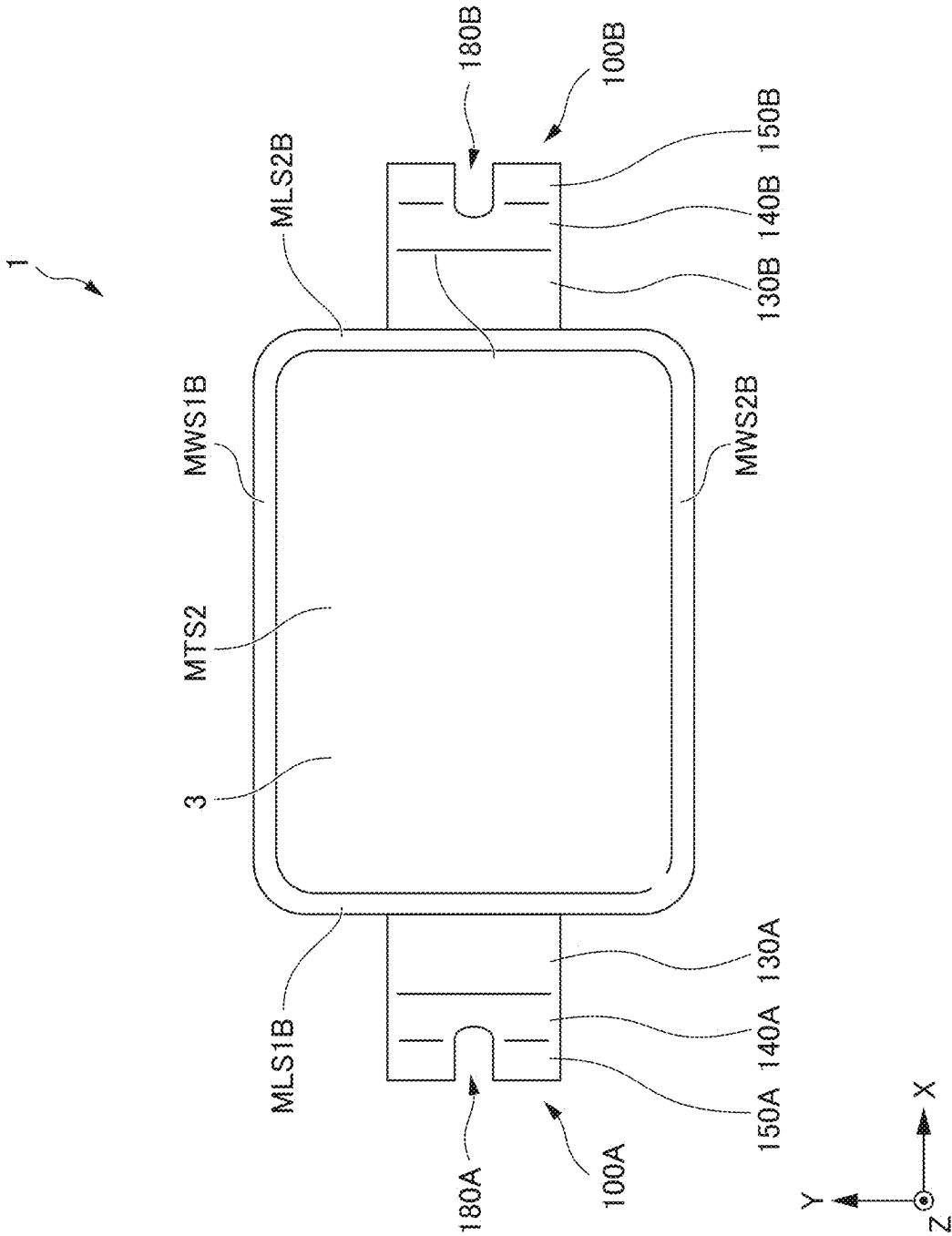


FIG. 4



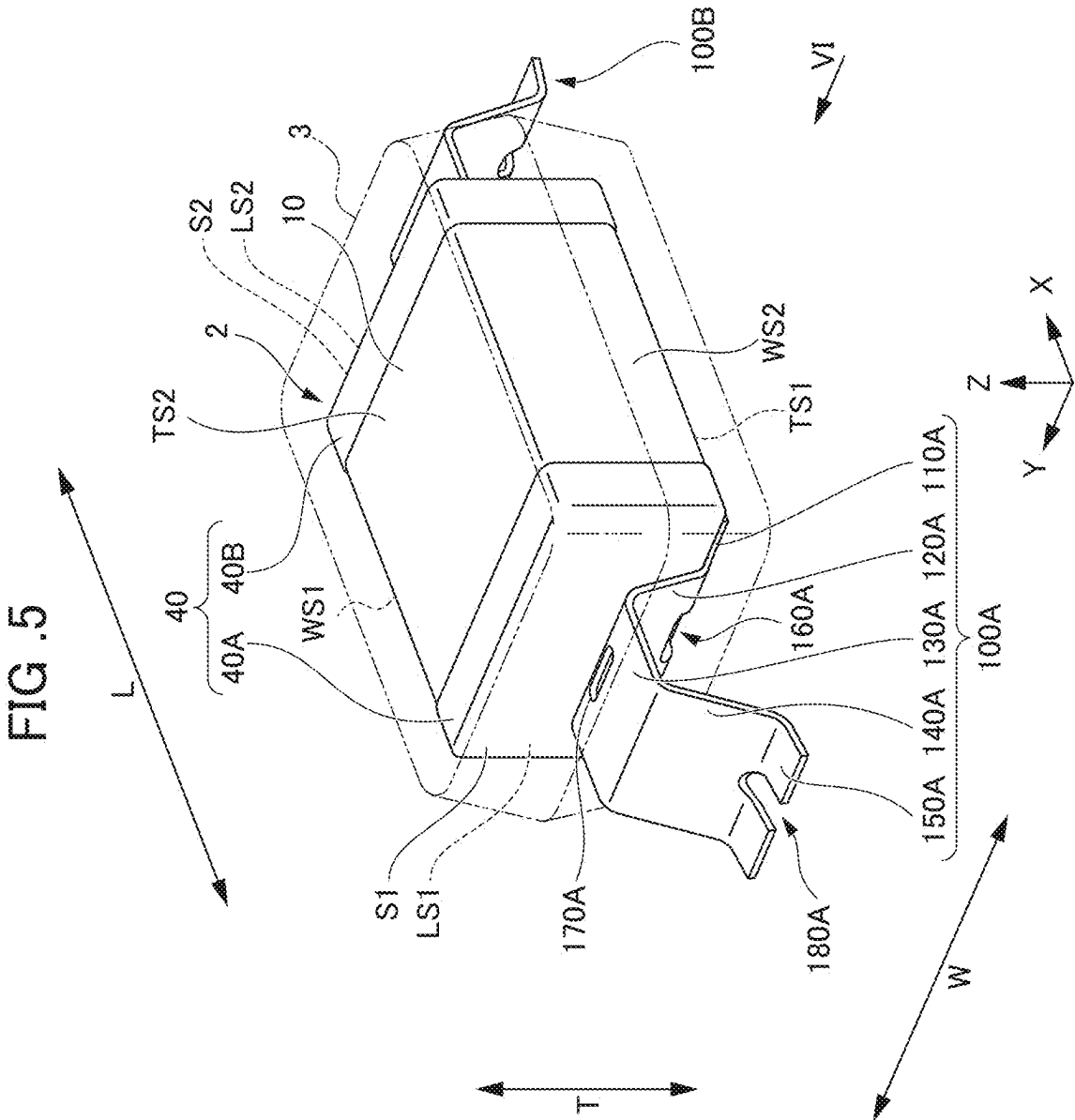


FIG. 6

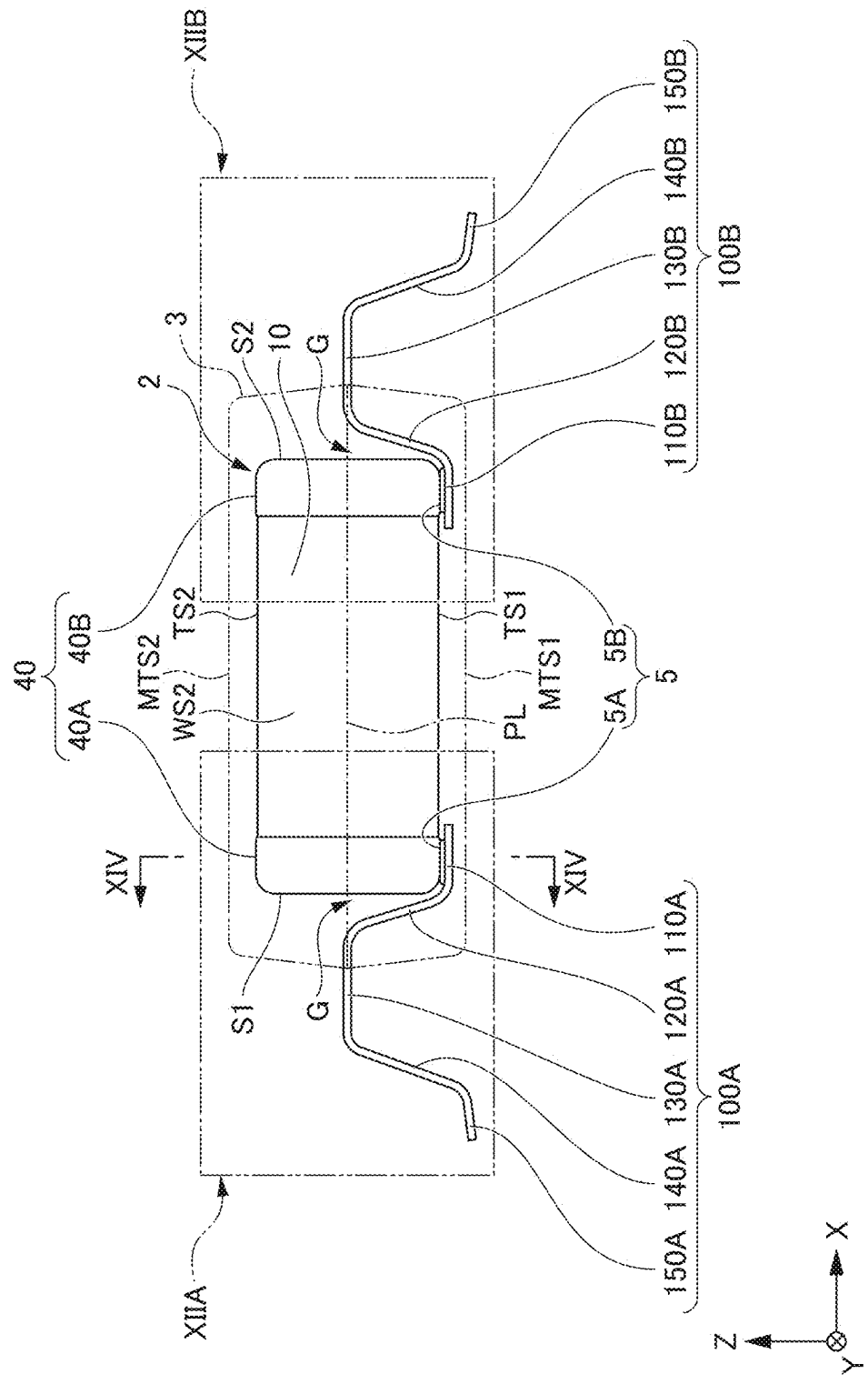
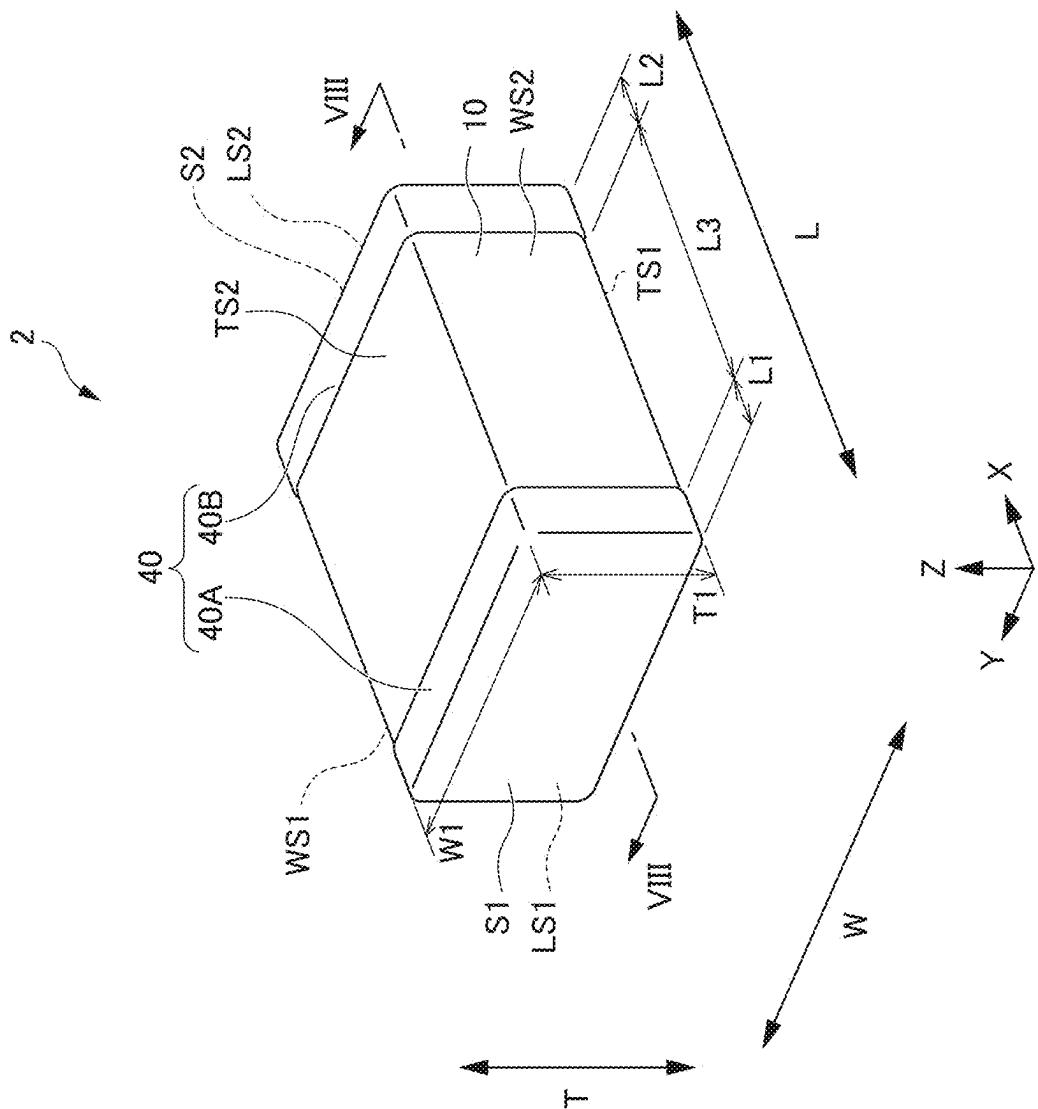
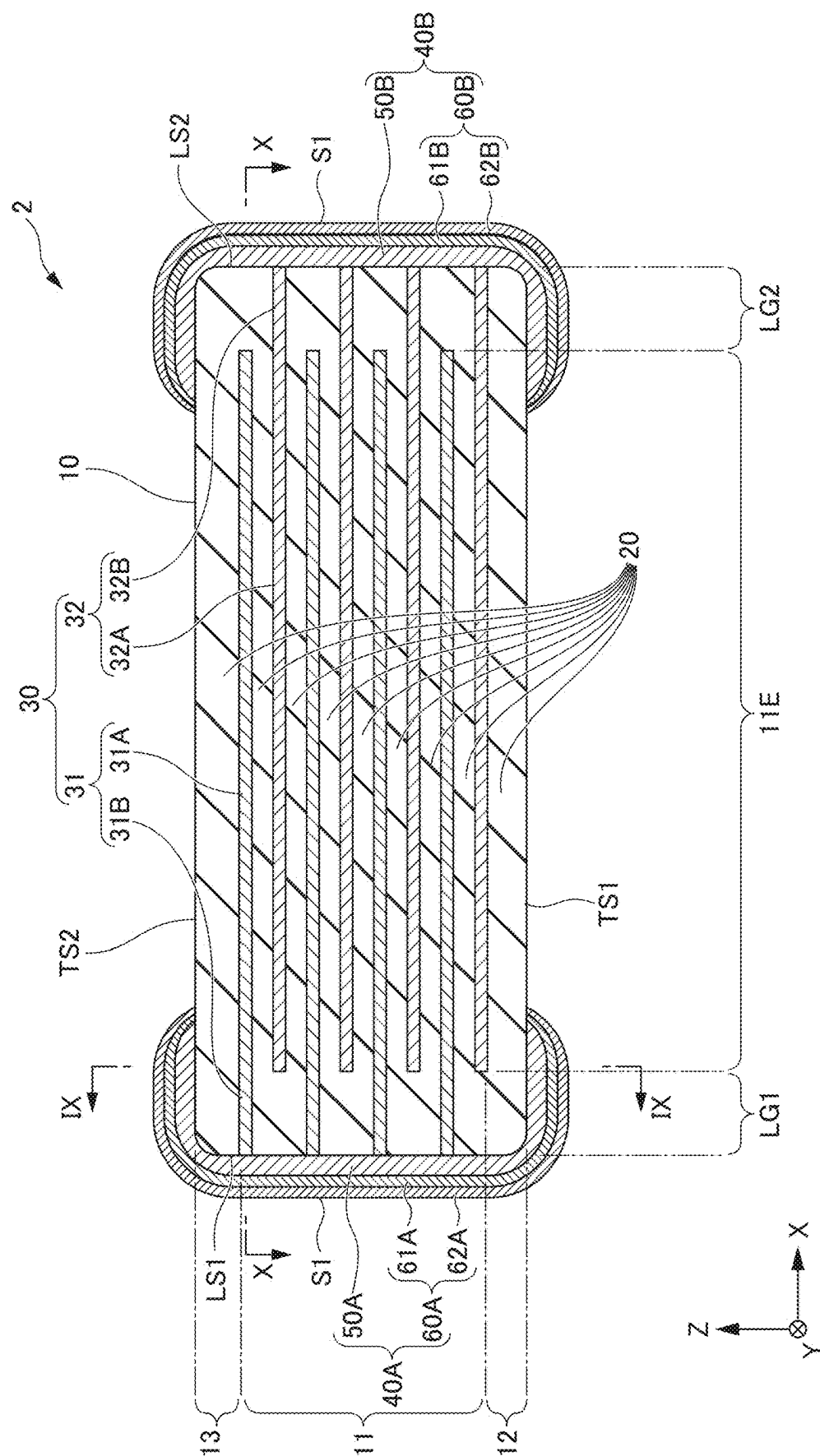


FIG. 7



8
9
10
11



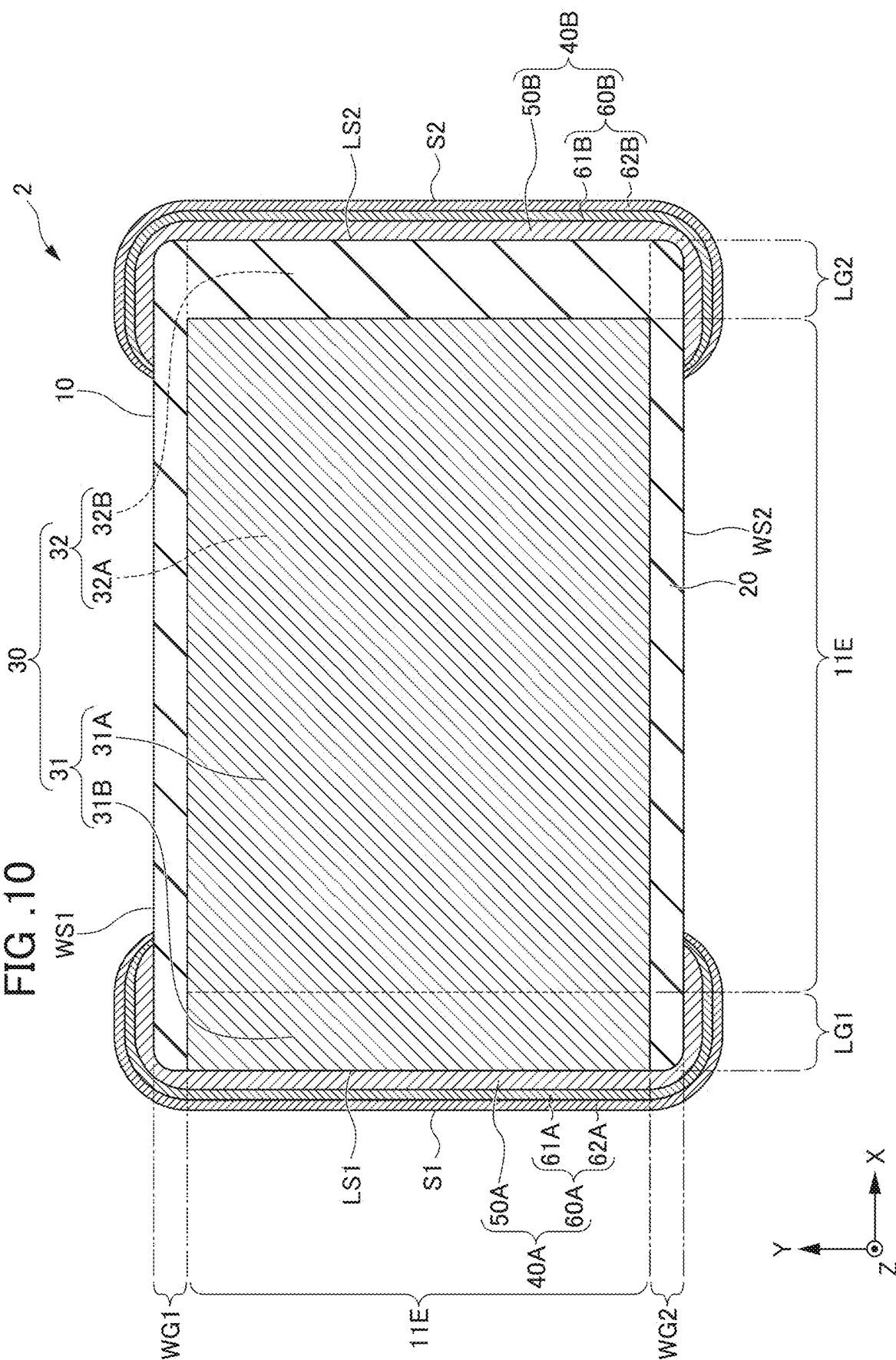


FIG. 11

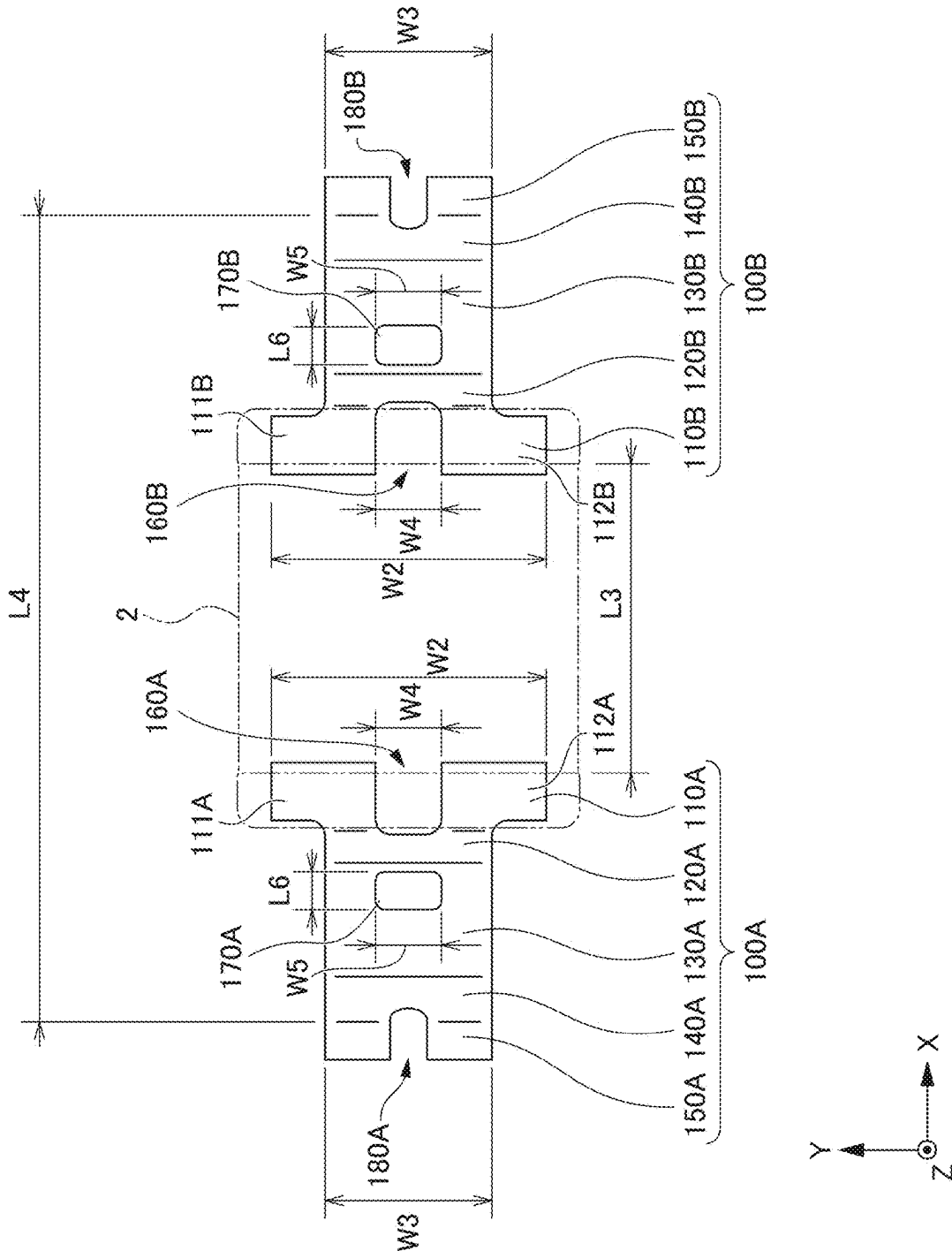


FIG. 12A

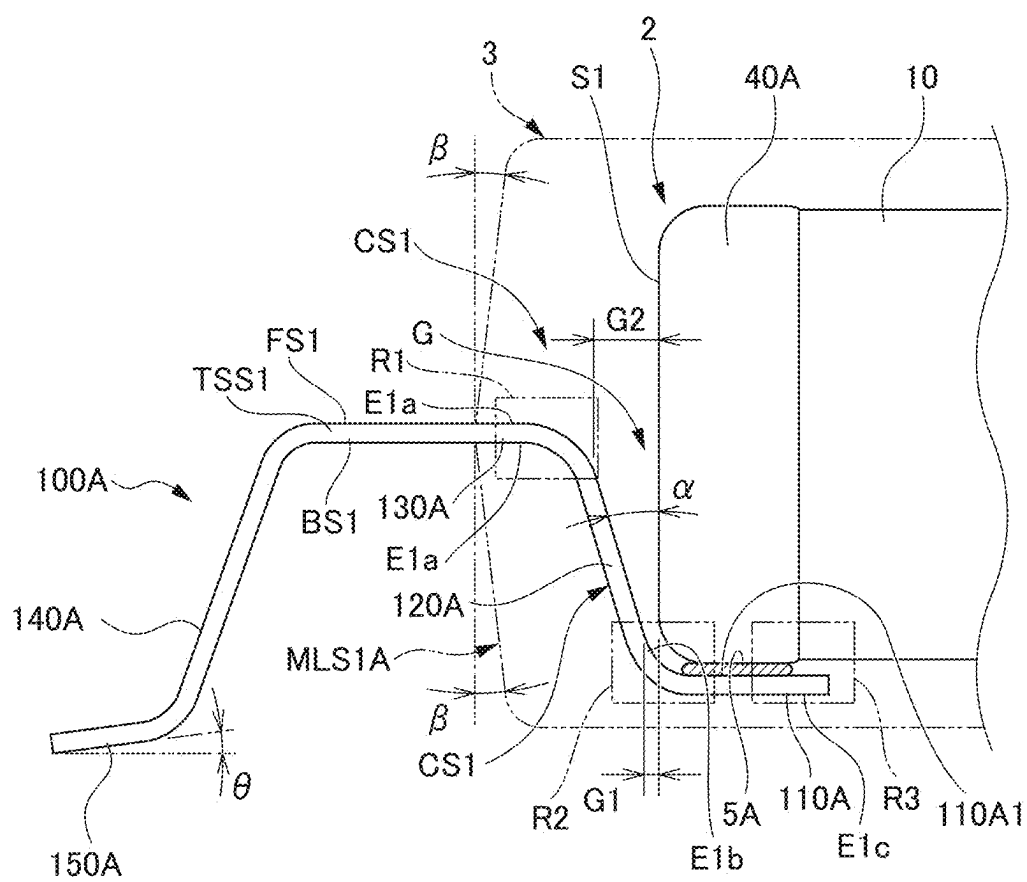


FIG. 12C

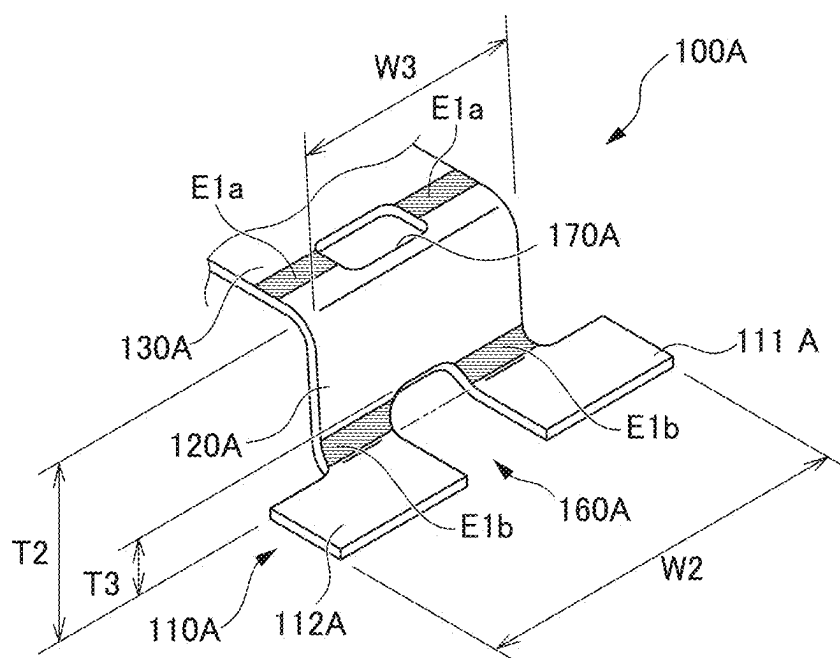


FIG. 12D

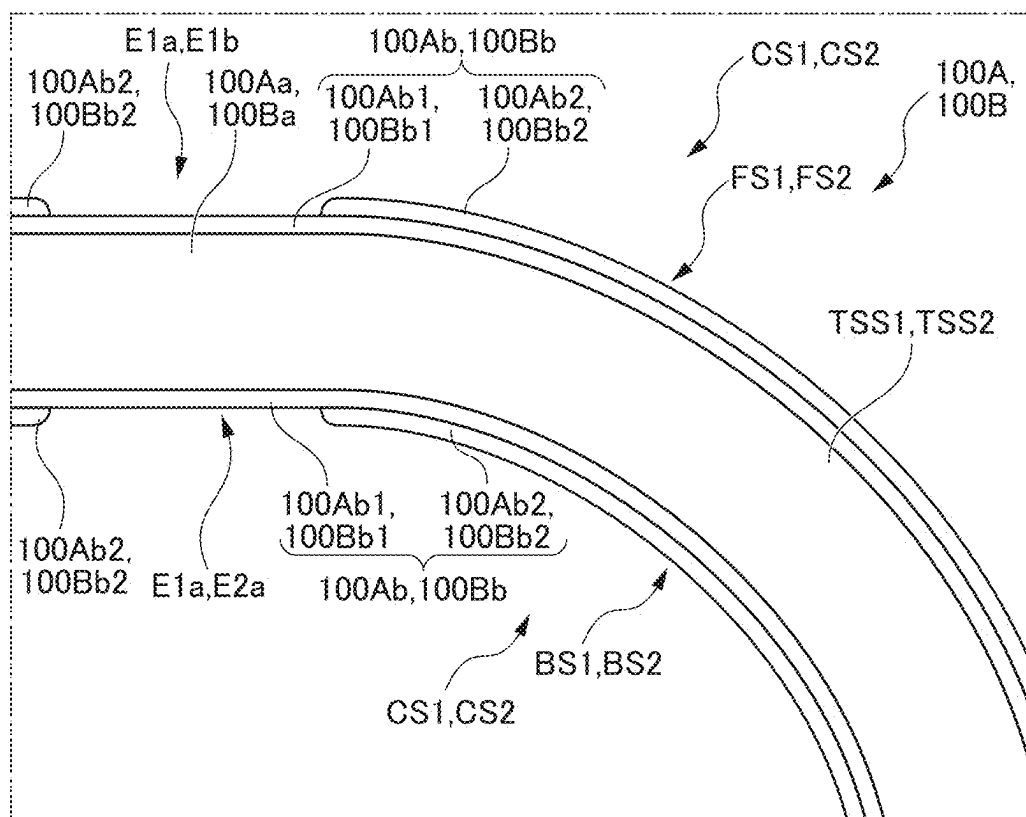


FIG. 12E

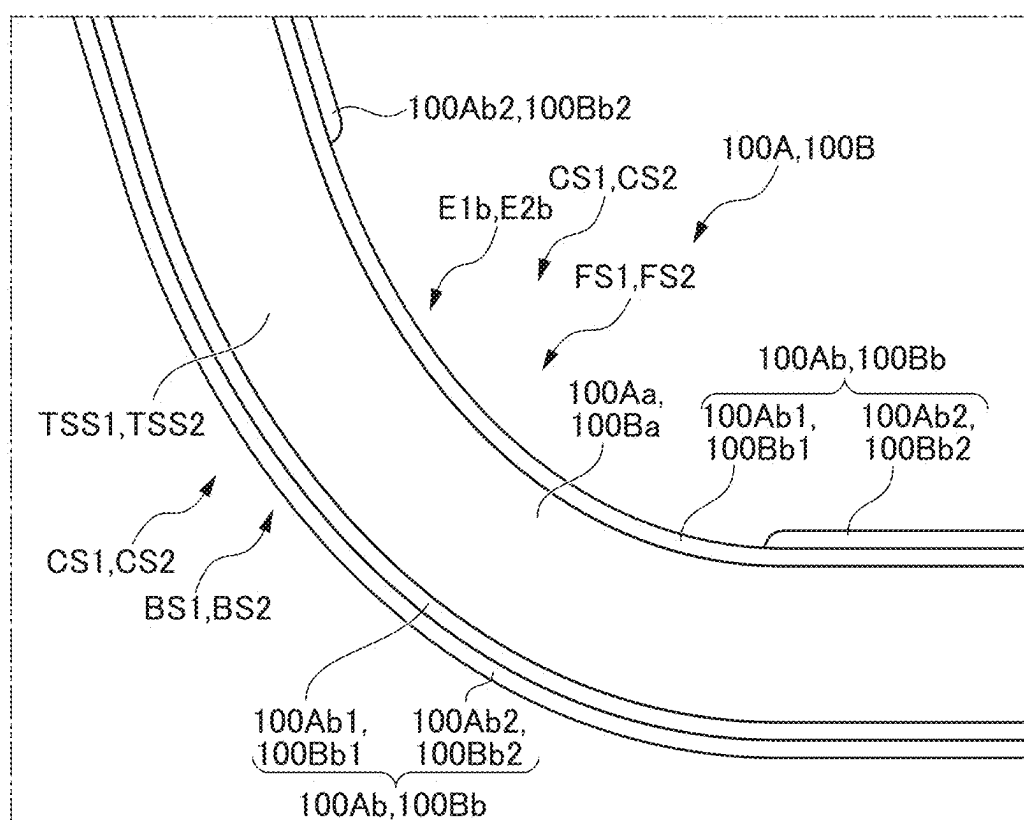


FIG. 12F

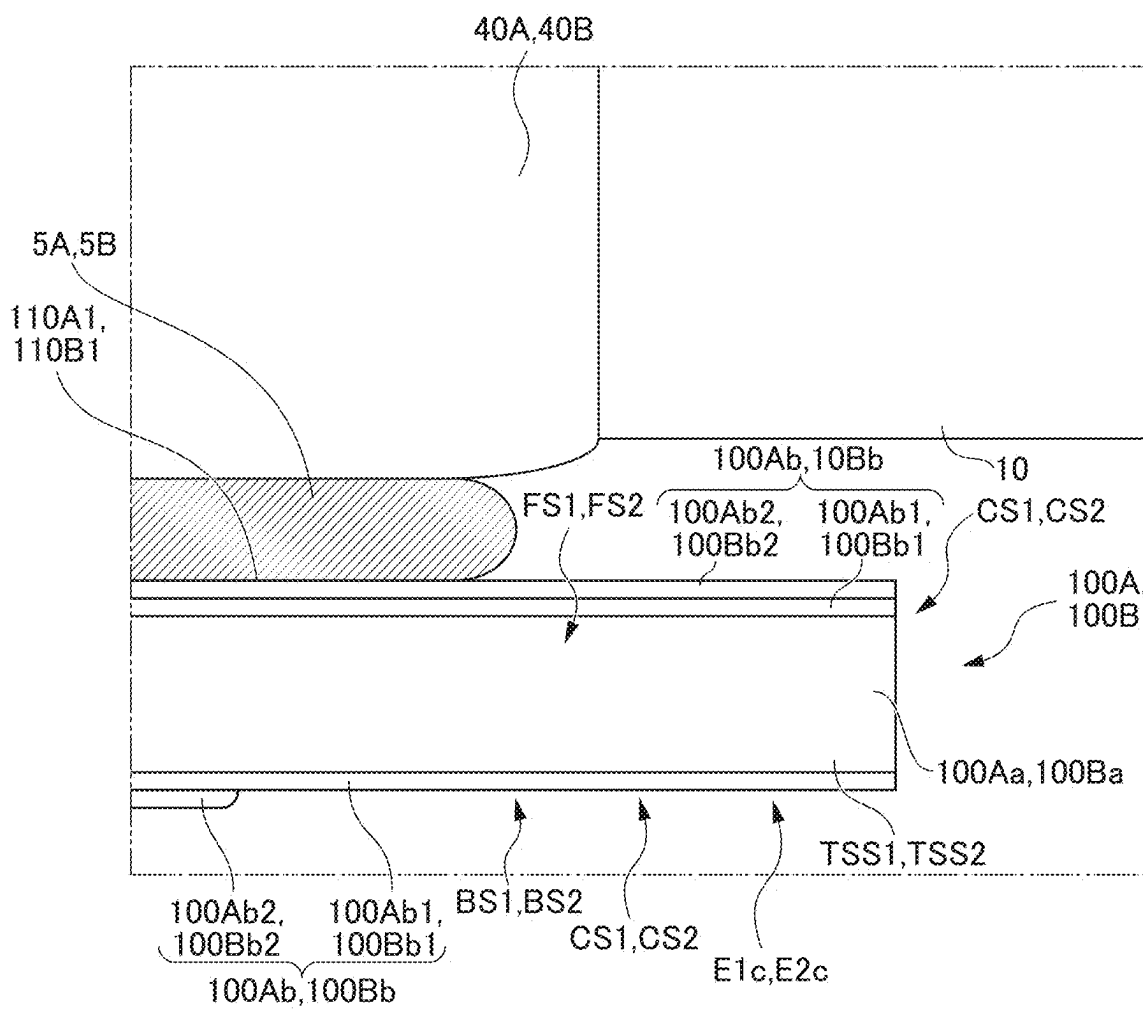


FIG. 13

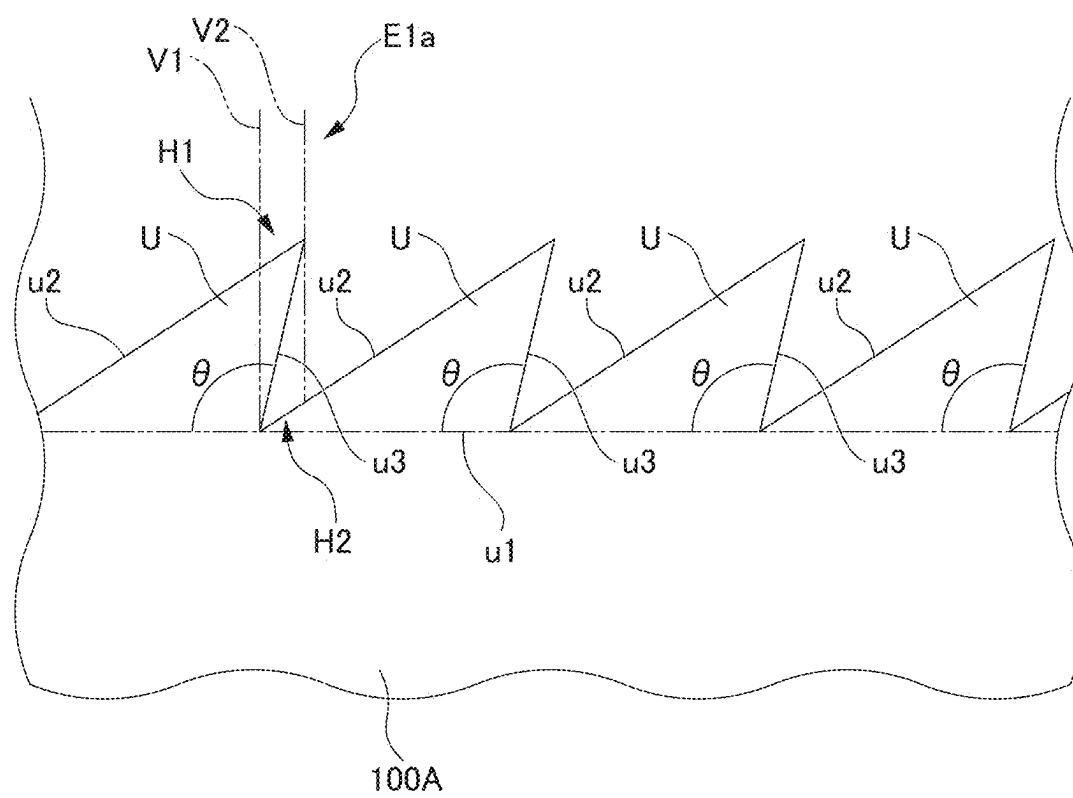


FIG. 14A

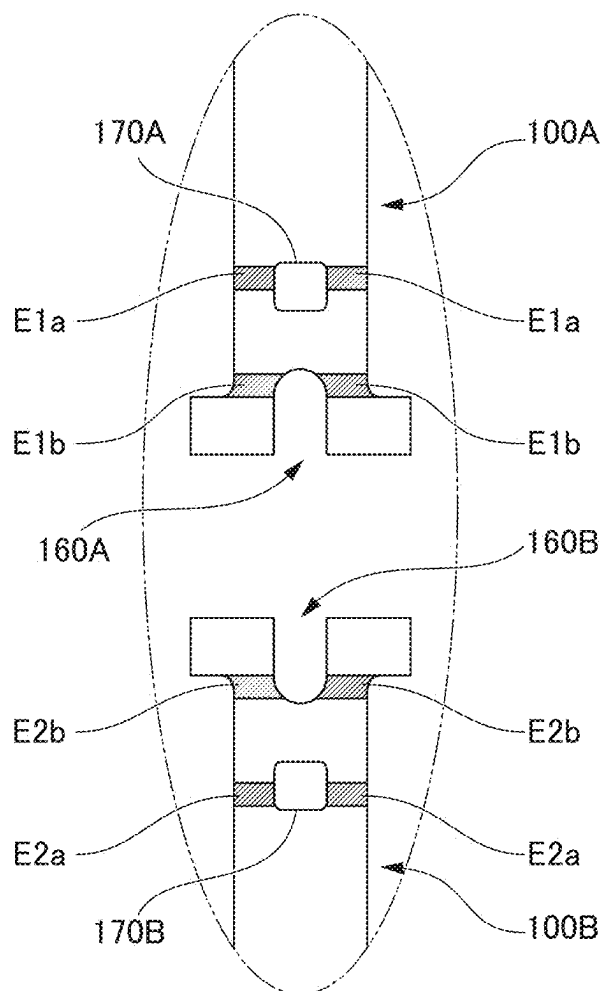


FIG. 15A

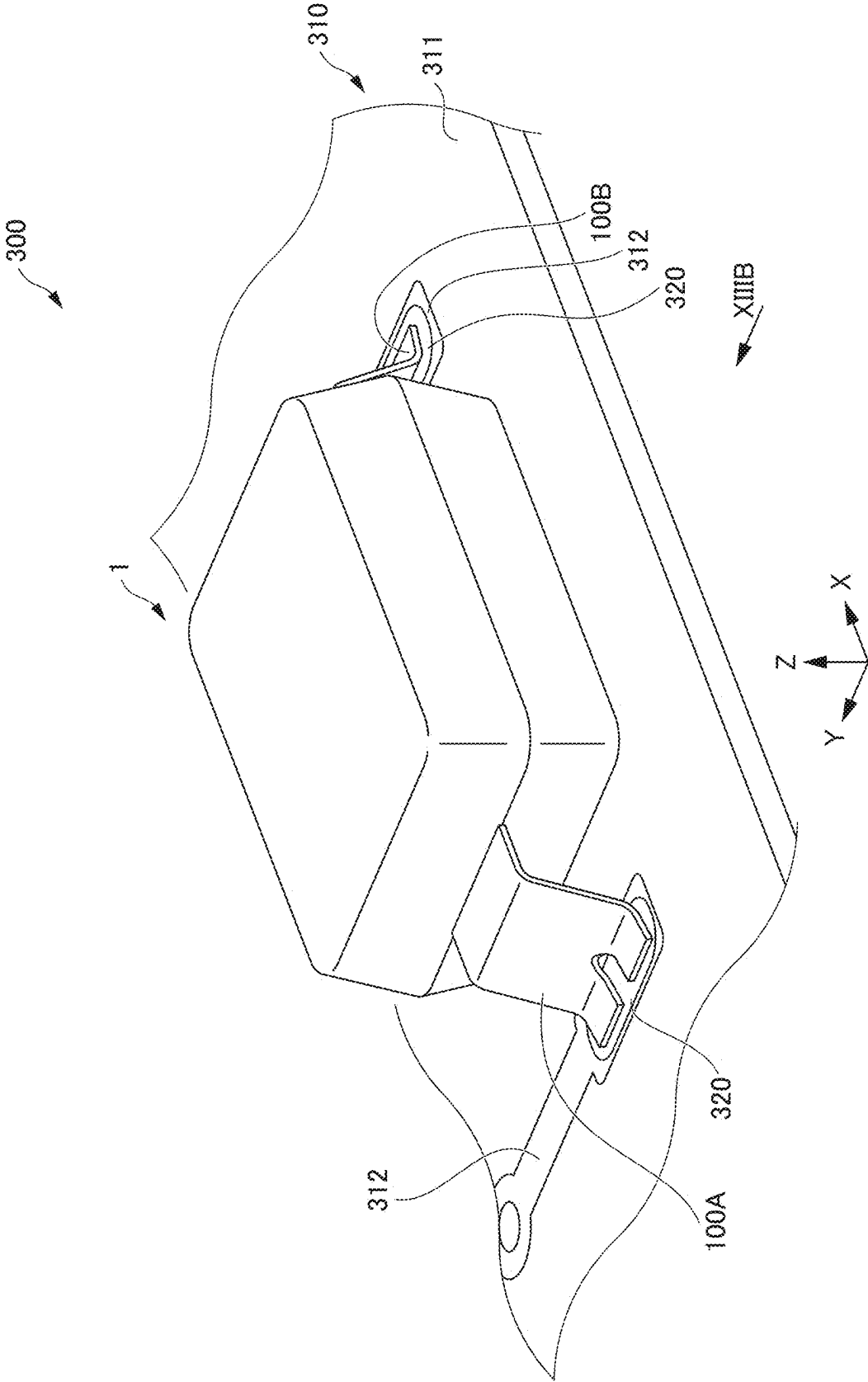


FIG. 16A

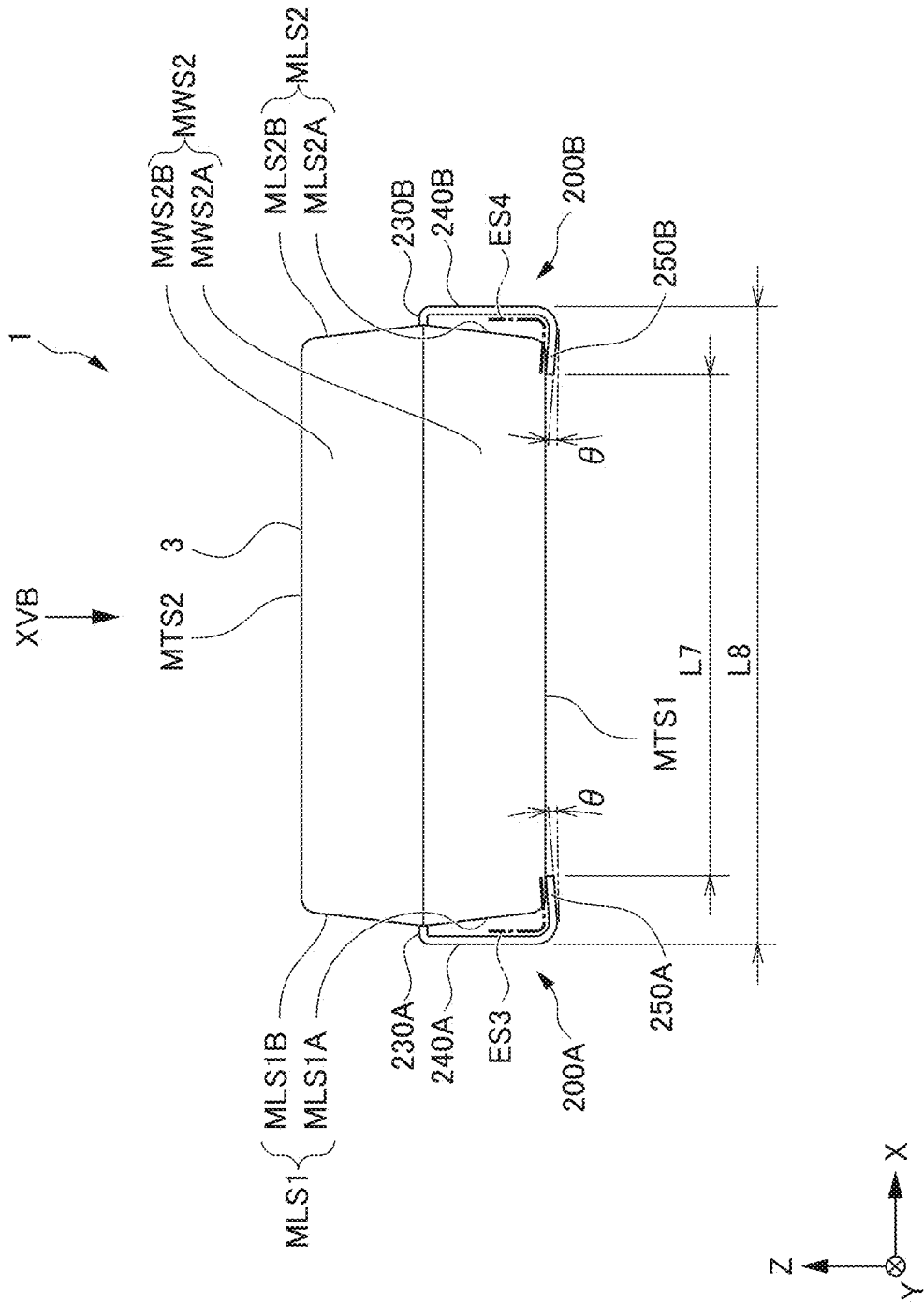


FIG. 16B

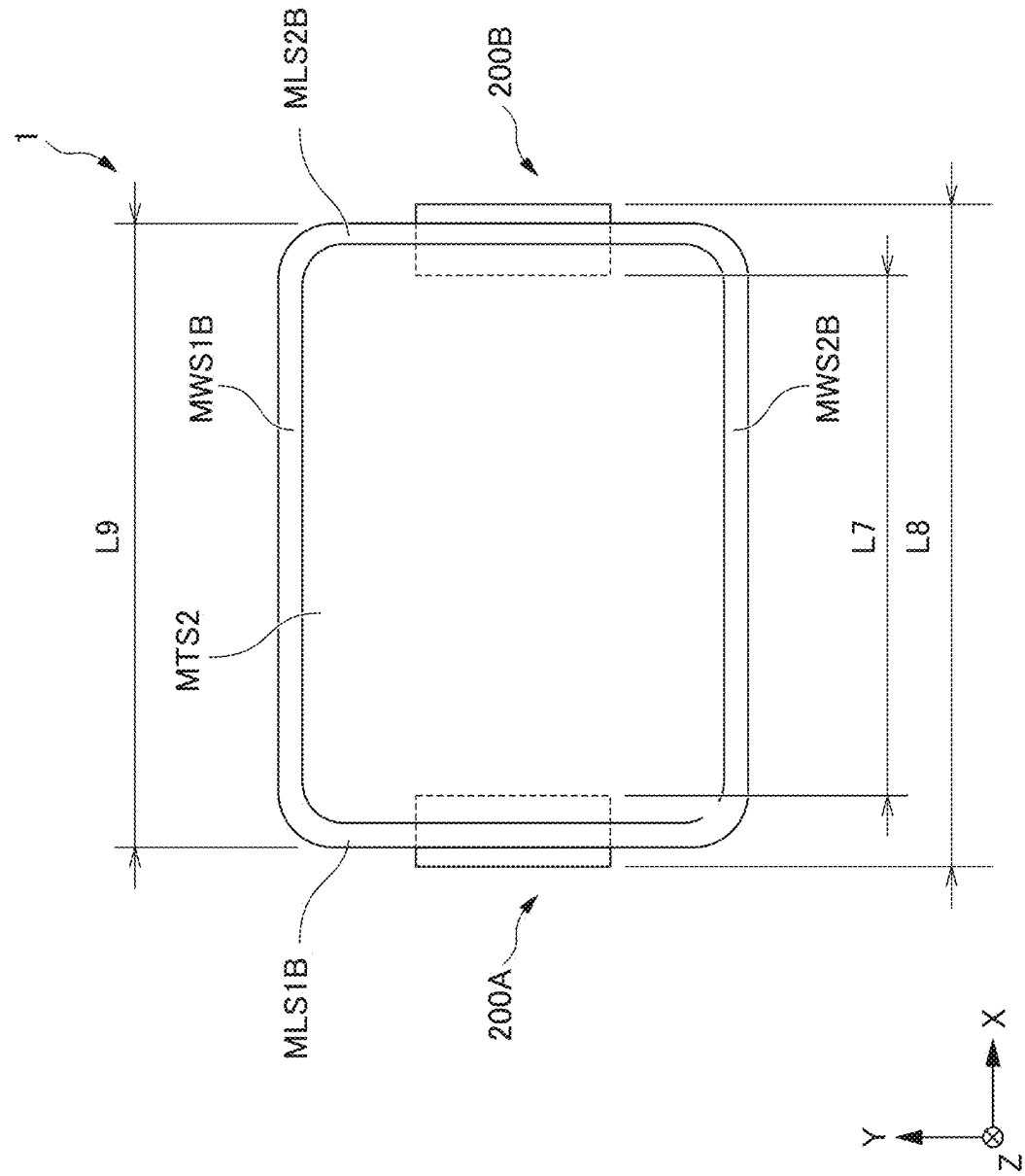


FIG. 17A

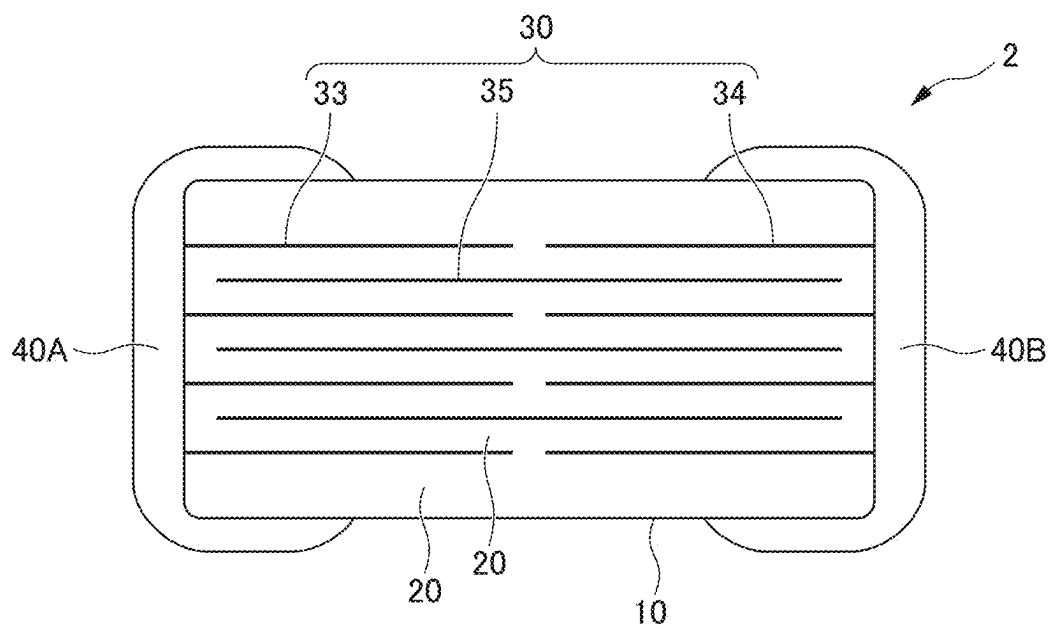


FIG. 17B

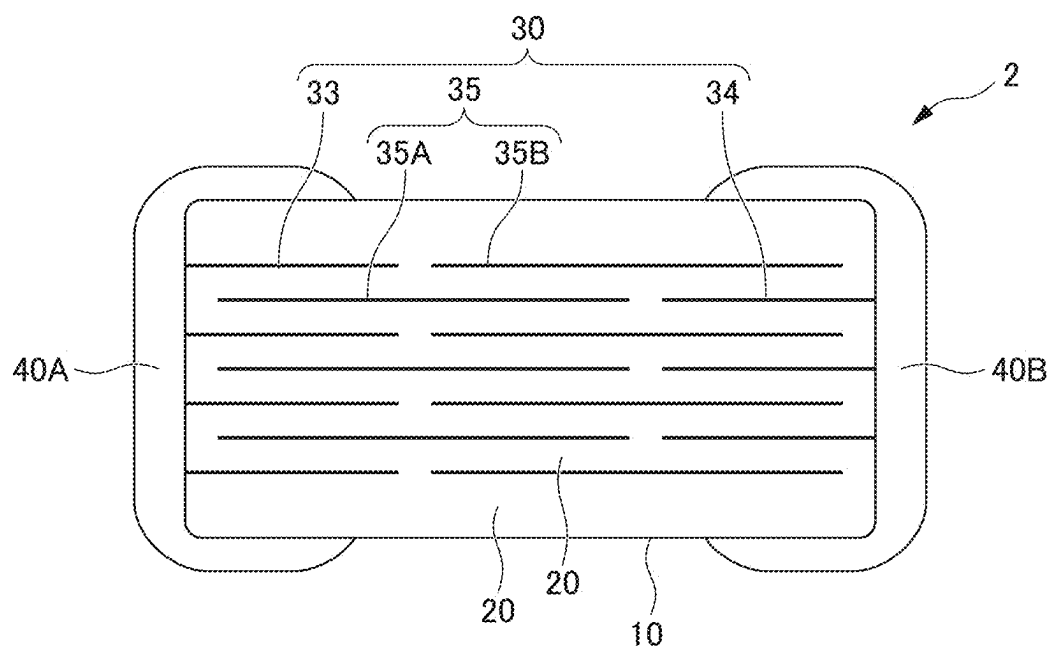
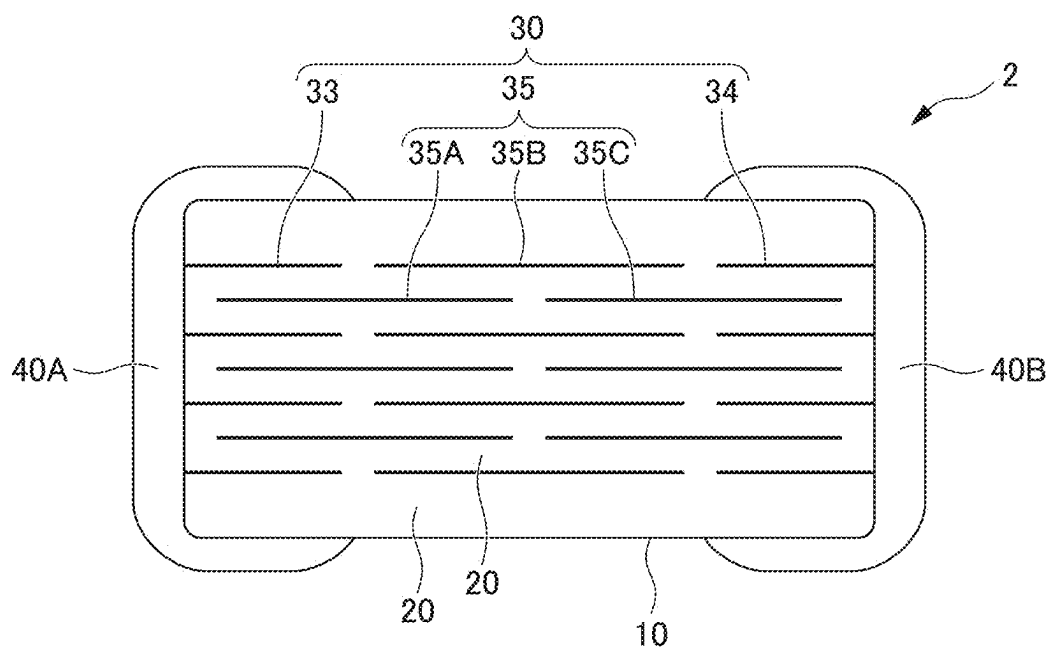


FIG. 17C



MULTILAYER CERAMIC ELECTRONIC COMPONENT

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to Japanese Patent Application No. 2022-202664 filed on Dec. 19, 2022 and is a Continuation Application of PCT Application No. PCT/JP2023/042188 filed on Nov. 24, 2023. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to multilayer ceramic electronic components.

2. Description of the Related Art

[0003] Conventionally, multilayer ceramic electronic components each covered with a resin functioning as an exterior material are known. In each of such multilayer ceramic electronic components, metal terminals each extending to the outside of the exterior material and external electrodes each provided on the surface of the multilayer ceramic electronic component main body are bonded to each other by a bonding material including a metal such as solder inside the exterior material.

[0004] Japanese Unexamined Patent Application, Publication No. 2019-145767 discloses a multilayer ceramic electronic component including a plating film on a surface of a frame functioning as a metal terminal. Since the plating film is provided on the metal terminal, the bonding property by the bonding material can be enhanced. However, since the surface of the plating film formed on the metal terminal is smooth, there is a possibility that peeling will occur at the interface between the metal terminal and the exterior material. In this case, external vapor may enter the interior of the exterior material, and the reliability of the electronic component may decrease.

SUMMARY OF THE INVENTION

[0005] Example embodiments of the present invention provide multilayer ceramic electronic components that each reduce or prevent peeling at an interface between a metal terminal and an exterior material and reduce or prevent a decrease in reliability of the electronic component.

[0006] A multilayer ceramic electronic component according to an example embodiment of the present invention includes a multilayer ceramic electronic component main body including a multilayer body including a plurality of ceramic layers, and a plurality of internal conductive layers that are each laminated on a corresponding one of the plurality of ceramic layers, a first main surface and a second main surface opposed to each other in a height direction, a first lateral surface and a second lateral surface opposed to each other in a width direction orthogonal or substantially orthogonal to the height direction, and a first end surface and a second end surface opposed to each other in a length direction orthogonal or substantially orthogonal to the height direction and the width direction, a first external electrode on the first end surface, and a second external electrode on the second end surface, a first metal terminal connected to the

first external electrode via a bonding material, a second metal terminal connected to the second external electrode via a bonding material, and an exterior material covering the multilayer ceramic electronic component main body, the bonding material, a portion of the first metal terminal, and a portion of the second metal terminal, in which the first metal terminal includes a first bonding surface bonded to the bonding material, and a first contact surface in contact with the exterior material and including a first outermost surface metal film on at least a portion of a surface of the first contact surface, the second metal terminal includes a second bonding surface bonded to the bonding material, and a second contact surface in contact with the exterior material and including a second outermost surface metal film on at least a portion of a surface of the second contact surface, the first contact surface includes a plurality of first exposed surfaces each including an exposed surface of a metal different from the first outermost surface metal film, the second contact surface includes a plurality of second exposed surfaces each including an exposed surface of a metal different from the second outermost surface metal film, the plurality of first exposed surfaces and the plurality of second exposed surfaces each include a plurality of protruding portions each having an undercut shape.

[0007] According to example embodiments of the present invention, it is possible to provide multilayer ceramic electronic components that are each able to reduce or prevent peeling at an interface between a metal terminal and an exterior material, and reduce or prevent a decrease in reliability of the electronic component.

[0008] The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is an external perspective view of a multilayer ceramic capacitor according to an example embodiment of the present invention.

[0010] FIG. 2 is an arrow view when the multilayer ceramic capacitor of FIG. 1 is viewed in the direction of the arrow II.

[0011] FIG. 3 is an arrow view when the multilayer ceramic capacitor of FIG. 2 is viewed in the direction of the arrow III.

[0012] FIG. 4 is an arrow view when the multilayer ceramic capacitor of FIG. 2 is viewed in the direction of the arrow IV.

[0013] FIG. 5 is a diagram corresponding to FIG. 1, and is a virtual perspective view for explaining an internal structure of a multilayer ceramic capacitor according to an example embodiment of the present invention.

[0014] FIG. 6 is a virtual arrow view when the multilayer ceramic capacitor of FIG. 5 is viewed in the direction of arrow VI.

[0015] FIG. 7 is an external perspective view showing the appearance of a multilayer ceramic capacitor main body according to an example embodiment of the present invention before being covered with an exterior material and before a metal terminal is attached.

[0016] FIG. 8 is a cross-sectional view taken along the line VIII-VIII of the multilayer ceramic capacitor main body of FIG. 7.

[0017] FIG. 9 is a cross-sectional view taken along the line IX-IX of the multilayer ceramic capacitor main body of FIG. 8.

[0018] FIG. 10 is a cross-sectional view taken along the line X-X of the multilayer ceramic capacitor main body of FIG. 8.

[0019] FIG. 11 is a view corresponding to FIG. 4, and is a view showing a metal terminal when an exterior material and a multilayer ceramic capacitor main body are excluded.

[0020] FIG. 12A is an enlarged view of a portion X11A of the multilayer ceramic capacitor shown in FIG. 6.

[0021] FIG. 12B is an enlarged view of a portion X11B of the multilayer ceramic capacitor shown in FIG. 6.

[0022] FIG. 12C is a partial external perspective view of a first metal terminal according to an example embodiment of the present invention.

[0023] FIG. 12D is an enlarged view of a portion R1 of the multilayer ceramic capacitor shown in FIG. 12A.

[0024] FIG. 12E is an enlarged view of a portion R2 of the multilayer ceramic capacitor shown in FIG. 12A.

[0025] FIG. 12F is an enlarged view of a portion R3 of the multilayer ceramic capacitor shown in FIG. 12A.

[0026] FIG. 13 is a cross-sectional view showing an example of a plurality of protruding portions provided on an exposed surface of the first metal terminal according to an example embodiment of the present invention.

[0027] FIG. 14A is a front view of the metal terminal before being folded.

[0028] FIG. 14B is a view showing an opposite surface of the metal terminal before being folded.

[0029] FIG. 15A is an external perspective view showing a mounting structure in which a multilayer ceramic capacitor according to an example embodiment of the present invention is mounted on a mounting substrate.

[0030] FIG. 15B is a view corresponding to FIG. 6, and is a virtual arrow view when the mounting structure of the multilayer ceramic capacitor of FIG. 15A is viewed from the direction of arrow XIII B.

[0031] FIG. 16A is a view showing a modified example of a multilayer ceramic capacitor according to an example embodiment of the present invention, and corresponds to FIG. 2.

[0032] FIG. 16B is an arrow view of the multilayer ceramic capacitor of FIG. 16A as viewed in the direction of arrow XV B.

[0033] FIG. 17A is a diagram showing a multilayer ceramic capacitor having a two-portion structure.

[0034] FIG. 17B is a diagram showing a multilayer layer ceramic capacitor having a three-portion structure.

[0035] FIG. 17C is a diagram showing a multilayer layer ceramic capacitor having a four-portion structure.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0036] Example embodiments of the present invention will be described in detail below with reference to the drawings.

[0037] Hereinafter, multilayer ceramic capacitors as multilayer ceramic electronic component according to example embodiments of the present invention will be described. FIG. 1 is an external perspective view of the multilayer ceramic capacitor 1. FIG. 2 is an arrow view when the multilayer ceramic capacitor 1 of FIG. 1 is viewed in the direction of the arrow II. FIG. 3 is an arrow view when the

multilayer ceramic capacitor 1 of FIG. 2 is viewed in the direction of the arrow III. FIG. 4 is an arrow view when the multilayer ceramic capacitor 1 of FIG. 2 is viewed in the direction of the arrow IV. FIG. 5 is a diagram corresponding to FIG. 1, and is a virtual perspective view for explaining an internal structure of the multilayer ceramic capacitor 1. FIG. 6 is a virtual view for explaining the internal structure of the multilayer ceramic capacitor 1, and is a virtual view when the multilayer ceramic capacitor 1 of FIG. 5 is viewed in the direction of the arrow VI.

[0038] The multilayer ceramic capacitor 1 includes a multilayer ceramic capacitor main body 2 defining and functioning as multilayer ceramic electronic component main body, a metal terminal 100, and an exterior material 3. Since the multilayer ceramic capacitor main body 2 is covered with the exterior material 3, it is not shown in FIGS. 1 to 4. FIGS. 5 and 6 show the multilayer ceramic capacitor main body 2.

[0039] The multilayer ceramic capacitor main body 2 will also be described with reference to FIGS. 7 to 10, in addition to FIGS. 5 and 6. FIG. 7 is an external perspective view showing the appearance of the multilayer ceramic capacitor main body 2 before being covered with the exterior material 3 and before the metal terminal 100 is attached. FIG. 8 is a cross-sectional view taken along the line VIII-VIII of the multilayer ceramic capacitor main body 2 of FIG. 7. FIG. 9 is a cross-sectional view taken along the line IX-IX of the multilayer ceramic capacitor main body 2 of FIG. 8. FIG. 10 is a cross-sectional view taken along the line X-X of the multilayer ceramic capacitor main body 2 of FIG. 8.

[0040] The multilayer ceramic capacitor main body 2 includes a multilayer body 10 and external electrodes 40.

[0041] FIGS. 5 to 10 each show an XYZ Cartesian coordinate system. As shown in FIGS. 5 and 7, the length directions L of the multilayer ceramic capacitor main body 2 and the multilayer body 10 correspond to the X direction. The width directions W of the multilayer ceramic capacitor main body 2 and the multilayer body 10 correspond to the Y direction. The height directions T of the multilayer ceramic capacitor main body 2 and the multilayer body 10 correspond to the Z direction. Here, the cross section shown in FIG. 8 is also referred to as a cross section LT. The cross section shown in FIG. 9 is also referred to as a cross section WT. The cross section shown in FIG. 10 is also referred to as a cross section LW. A similar XYZ Cartesian coordinate system is also shown in FIGS. 1 to 4, 11, and 15A to 16B.

[0042] As shown in FIGS. 5 to 10, the multilayer body 10 includes a first main surface TS1 and a second main surface TS2 which oppose each other in the height direction T, a first lateral surface WS1 and a second lateral surface WS2 which oppose each other in the width direction W orthogonal or substantially orthogonal to the height direction T, and a first end surface LS1 and a second end surface LS2 which oppose each other in the length direction L orthogonal or substantially orthogonal to the height direction T and the width direction W.

[0043] As shown in FIG. 7, the multilayer body 10 includes a rectangular or substantially rectangular shape. The dimension of the multilayer body 10 in the length direction L is not necessarily longer than the dimension of the width direction W. The multilayer body 10 preferably includes rounded corner portions and rounded ridge portions. The corner portions are portions where the three surfaces of the multilayer body intersect, and the ridge

portions are portions where the two surfaces of the multilayer body intersect. In addition, unevenness or the like may be provided on a portion or an entirety of the surface of the multilayer body 10.

[0044] The dimension of the multilayer body 10 is not particularly limited. However, when the dimension in the length direction L of the multilayer body 10 is defined as L, L is, for example, preferably about 0.2 mm or more and about 10 mm or less. When the dimension in the height direction T of the multilayer body 10 is defined as T, T is, for example, preferably about 0.1 mm or more and about 10 mm or less. Furthermore, when the dimension in the width direction W of the multilayer body 10 is defined as W, W is, for example, preferably about 0.1 mm or more and about 10 mm or less.

[0045] As shown in FIGS. 8 and 9, the multilayer body 10 includes an inner layer portion 11, and a first main surface-side outer layer portion 12 and a second main surface-side outer layer portion 13 sandwiching the inner layer portion 11 in the height direction T. The inner layer portion 11 may also be referred to as an active layer portion.

[0046] The inner layer portion 11 includes a plurality of dielectric layers 20 defining and functioning as a plurality of ceramic layers, and a plurality of internal electrode layers 30 defining and functioning as a plurality of inner conductive layers. The inner layer portion 11 includes internal electrode layers, in the height direction T, from the internal electrode layer 30 located closest to the first main surface TS1 to the internal electrode layer 30 located closest to the second main surface TS2. In the inner layer portion 11, the plurality of internal electrode layers 30 are opposed to each other with the dielectric layer 20 interposed therebetween. The inner layer portion 11 is a portion that generates a capacitance, and thus substantially defines and functions as a capacitor.

[0047] The plurality of dielectric layers 20 are made of a dielectric material. For example, the dielectric material may be a dielectric ceramic including a component such as BaTiO_3 , CaTiO_3 , SrTiO_3 , or CaZrO_3 . Furthermore, the dielectric material may be obtained by adding a second component such as a Mn compound, an Fe compound, a Cr compound, a Co compound, or a Ni compound to the main component.

[0048] The dielectric layers 20 each preferably have a thickness of, for example, about 0.5 μm or more and about 72 μm or less. The number of the dielectric layers 20 to be stacked (laminated) is, for example, preferably ten or more and 700 or less. The number of the dielectric layers 20 refers to the total number of dielectric layers in the inner layer portion 11, and dielectric layers in the first main surface-side outer layer portion 12 and the second main surface-side outer layer portion 13.

[0049] The plurality of internal electrode layers 30 (internal conductive layer 30) include a plurality of first internal electrode layers 31 (first internal conductive layer 31) and a plurality of second internal electrode layers 32 (second internal conductive layer 32). The plurality of first internal electrode layers 31 are provided on the plurality of dielectric layers 20. The plurality of second internal electrode layers 32 are provided on the plurality of dielectric layers 20. The plurality of first internal electrode layers 31 and the plurality of second internal electrode layers 32 are alternately provided in the height direction T of the multilayer body 10 with the dielectric layers 20 interposed therebetween. The first

internal electrode layers 31 and the second internal electrode layers 32 sandwich the dielectric layers 20.

[0050] The first internal electrode layer 31 includes a first counter portion 31A that is opposed to the second internal electrode layer 32, and a first extension portion 31B extending from the first counter portion 31A toward the first end surface LS1. The first extension portion 31B is exposed at the first end surface LS1.

[0051] The second internal electrode layer 32 includes a second counter portion 32A that is opposed to the first internal electrode layer 31, and a second extension portion 32B extending from the second counter portion 32A toward the second end surface LS2. The second extension portion 32B is exposed at the second end surface LS2.

[0052] In the present example embodiment, the first counter portion 31A and the second counter portion 32A are opposed to each other with the dielectric layer 20 interposed therebetween, such that a capacitance is generated, so as to provide the characteristics of a capacitor.

[0053] The shapes of the first counter portion 31A and the second counter portion 32A are not particularly limited. However, they are preferably rectangular or substantially rectangular. However, the corner portions of the rectangular shape may be rounded or slanted. The shapes of the first extension portion 31B and the second extension portion 32B are not particularly limited. However, they are preferably rectangular or substantially rectangular. However, the corner portions of the rectangular shape may be rounded or slanted.

[0054] The dimension in the width direction W of the first counter portion 31A and the dimension in the width direction W of the first extension portion 31B may have the same or substantially the same dimensions, or one of them may have a smaller dimension. The dimension in the width direction W of the second counter portion 32A and the dimension in the width direction W of the second extension portion 32B may have the same or substantially the same dimension, or one of them may have a narrower dimension.

[0055] The first internal electrode layer 31 and the second internal electrode layer 32 are each made of a metal such as, for example, Ni, Cu, Ag, Pd, or Au, or a suitable electrically conductive material such as an alloy including at least one of these metals. In a case in which an alloy is used, the first internal electrode layer 31 and the second internal electrode layer 32 may be made of, for example, a Ag—Pd alloy.

[0056] The thickness of each of the first internal electrode layer 31 and the second internal electrode layer 32 is preferably, for example, about 0.2 μm or more and about 3.0 μm or less. The total number of the first internal electrode layers 31 and the second internal electrode layers 32 is, for example, preferably five or more and 350 or less.

[0057] The first main surface-side outer layer portion 12 is located adjacent to the first main surface TS1 of the multilayer body 10. The first main surface-side outer layer portion 12 is an assembly including a plurality of dielectric layers 20 defining and functioning as ceramic layers located between the first main surface TS1 and the internal electrode layer 30 closest to the first main surface TS1. In other words, the first main surface-side outer layer portion 12 includes a plurality of dielectric layers 20 located between the first main surface TS1 and the internal electrode layer 30 located closest to the first main surface TS1 among the plurality of internal electrode layers 30. The dielectric layers 20 in the first main surface-side outer layer portion 12 may be the same as the dielectric layers 20 in the inner layer portion 11.

[0058] The second main surface-side outer layer portion 13 is located adjacent to the second main surface TS2 of the multilayer body 10. The second main surface-side outer layer portion 13 is an assembly including a plurality of dielectric layers 20 located between the second main surface TS2 and the internal electrode layer 30 closest to the second main surface TS2. In other words, the second main surface-side outer layer portion 13 includes a plurality of dielectric layers 20 located between the second main surface TS2 and the internal electrode layer 30 located closest to the second main surface TS2 among the plurality of internal electrode layers 30. The dielectric layers 20 in the second main surface-side outer layer portion 13 may be the same as the dielectric layers 20 in the inner layer portion 11.

[0059] As described above, the multilayer body 10 includes the plurality of dielectric layers 20 and the plurality of internal electrode layers 30 laminated on the dielectric layers 20. That is, the multilayer ceramic capacitor 1 includes the multilayer body 10 including the dielectric layers 20 and the internal electrode layers 30 alternately laminated therein.

[0060] The multilayer body 10 includes a counter electrode portion 11E. The counter electrode portion 11E refers to a portion where a first counter portion 31A of each of the first internal electrode layers 31 and a second counter portion 32A of each of the second internal electrode layers 32 are opposed to each other. The counter electrode portion 11E defines and functions as a portion of the inner layer portion 11. FIG. 8 shows the range of the counter electrode portion 11E in the length direction L. FIG. 9 shows the range of the counter electrode portion 11E in the width direction W. FIG. 10 shows the ranges of the width direction W and the length direction L of the counter electrode portion 11E. The counter electrode portion 11E also referred to as a capacitor active portion.

[0061] The multilayer body 10 includes a lateral surface-side outer layer portion. The lateral surface-side outer layer portion includes a first lateral surface-side outer layer portion WG1 and a second lateral surface-side outer layer portion WG2. The first lateral surface-side outer layer portion WG1 includes the dielectric layers 20 located between the counter electrode portion 11E and the first lateral surface WS1. The second lateral surface-side outer layer portion WG2 includes the dielectric layers 20 located between the counter electrode portion 11E and the second lateral surface WS2. FIG. 9 and FIG. 10 each show the ranges in the width direction W of the first lateral surface-side outer layer portion WG1 and the second lateral surface-side outer layer portion WG2. The first lateral surface-side outer layer portion WG1 and the second lateral surface-side outer layer portion WG2 are also referred to as W gaps or side gaps.

[0062] The multilayer body 10 includes an end surface-side outer layer portion. The end surface-side outer layer portion includes a first end surface-side outer layer portion LG1 and a second end surface-side outer layer portion LG2. The first end surface-side outer layer portion LG1 includes the dielectric layers 20 located between the counter electrode portion 11E and the first end surface LS1, and the first extension portions 31B. The second end surface-side outer layer portion LG2 includes the dielectric layers 20 located between the counter electrode portion 11E and the second end surface LS2, and the second extension portion 32B. FIG. 8 and FIG. 10 each show the ranges in the length direction L of the first end surface-side outer layer portion LG1 and

the second end surface-side outer layer portion LG2. The first end surface-side outer layer portion LG1 and the second end surface-side outer layer portion LG2 are also referred to as L gaps or end gaps.

[0063] The external electrode 40 includes a first external electrode 40A provided on the first end surface LS1 and a second external electrode 40B provided on the second end surface LS2.

[0064] The first external electrode 40A is provided at least on a portion of the first main surface TS1 adjacent to the first end surface LS1. The first external electrode 40A is preferably provided at least on the first end surface LS1 and a portion on the first main surface TS1. In the present example embodiment of the present invention, the first external electrode 40A is provided on the first end surface LS1, a portion of the first main surface TS1, a portion of the second main surface TS2, a portion of the first lateral surface WS1, and a portion of the second lateral surface WS2. Furthermore, in the present example embodiment of the present invention, the first external electrode 40A is connected to the first internal electrode layers 31 on the first end surface LS1. Furthermore, for example, the first external electrode 40A may extend from the first end surface LS1 to a portion of the first main surface TS1. In other words, the cross-section of the first external electrode 40A may have an L shape (not shown). The portion provided on the first main surface TS1 of the first external electrode 40A is connected to a first metal terminal 100A described later.

[0065] The length L1 in the length direction L of the first external electrode 40A provided on the first main surface TS1 is, for example, preferably about 10% or more and about 40% or less (for example, about 20 μm or more and about 4000 μm or less) of the dimension L of the multilayer body. In a case in which the first external electrode 40A is provided on the second main surface TS2, the first lateral surface WS1, and the second lateral surface WS2, the length L1 in the length direction L of the first external electrode 40A provided on these surfaces is also, for example, preferably about 10% or more and about 40% or less (for example, about 20 μm or more and about 4000 μm or less) of the dimension L of the multilayer body.

[0066] The length W1 in the width direction W of the first external electrode 40A provided on the first main surface TS1 is preferably a dimension equal or substantially equal to the dimension W of the multilayer body 10 (for example, about 0.1 mm or more and about 10 mm or less). In a case in which the first external electrode 40A is also provided on the second main surface TS2, the length W1 in the width direction W of the first external electrode 40A provided on the second main surface TS2 is preferably a dimension equal or substantially equal to the dimension W of the multilayer body 10 (for example, about 0.1 mm or more and about 10 mm or less). Furthermore, in a case in which the first external electrode 40A is provided on at least one surface of the first lateral surface WS1 or the second lateral surface WS2, the length T1 in the height direction T of the first external electrode 40A provided on this portion is preferably a dimension equal or substantially equal to the dimension T of the multilayer body 10 (for example, about 0.1 mm or more and about 10 mm or less).

[0067] The second external electrode 40B is provided at least on a portion of the first main surface TS1 adjacent to the second end surface LS2. The second external electrode 40B is preferably provided at least on the second end surface

LS2 and a portion on the first main surface TS1. In the present example embodiment of the present invention, the second external electrode 40B is provided on the second end surface LS2, a portion of the first main surface TS1, a portion of the second main surface TS2, a portion of the first lateral surface WS1, and a portion of the second lateral surface WS2. Further, in the present example embodiment, the second external electrode 40B is connected to the second internal electrode layers 32 on the second end surface LS2. Furthermore, for example, the second external electrode 40B may extend from the second end surface LS2 to a portion of the first main surface TS1. In other words, the cross-section of the second external electrode 40B may have an L shape (not shown). The portion provided on the first main surface TS1 of the second external electrode 40B is connected to a second metal terminal 100B described later via a bonding material.

[0068] The length L2 in the length direction L of the second external electrode 40B provided on the first main surface TS1 is, for example, preferably about 10% or more and about 40% or less (for example, about 20 μm or more and about 4000 μm or less) of the dimension L of the multilayer body. In a case in which the second external electrode 40B is provided on the second main surface TS2, the first lateral surface WS1, and the second lateral surface WS2, the length L2 in the length direction L of the second external electrode 40B provided on these surfaces is also, for example, preferably about 10% or more and about 40% or less (for example, about 20 μm or more and about 4000 μm or less) of the dimension L of the multilayer body.

[0069] The length W1 in the width direction W of the second external electrode 40B provided on the first main surface TS1 is preferably a dimension equal or substantially equal to the dimension W of the multilayer body 10 (for example, about 0.1 mm or more and about 10 mm or less). In a case in which the second external electrode 40B is also provided on the second main surface TS2, the length W1 in the width direction W of the second external electrode 40B provided on the second main surface TS2 is preferably a dimension equal or substantially equal to the dimension W of the multilayer body 10 (for example, about 0.1 mm or more and about 10 mm or less). Furthermore, in a case in which the second external electrode 40B is provided on at least one surface of the first lateral surface WS1 or the second lateral surface WS2, the length T1 in the height direction T of the second external electrode 40B provided on this portion is preferably a dimension equal or substantially equal to the dimension T of the multilayer body 10 (for example, about 0.1 mm or more and about 10 mm or less).

[0070] As shown in FIG. 7, in the present example embodiment of the present invention, the length L3 in the length direction L of the portion of the surface of the multilayer body 10 exposed from the external electrode 40 is, for example, preferably about 20% or more and about 80% or less (for example, about 40 μm or more and about 8000 μm or less) of the dimension L of the multilayer body. In other words, the separation distance L3 between the first external electrode 40A and the second external electrode 40B is, for example, preferably about 20% or more and about 80% or less (for example, about 40 μm or more and about 8000 μm or less) of the dimension L of the multilayer body.

[0071] As described above, in the multilayer body 10, the capacitance is generated by the first counter portions 31A of

the first internal electrode layers 31 and the second counter portions 32A of the second internal electrode layers 32 being opposed to each other with the dielectric layers 20 interposed therebetween. Therefore, the characteristics of the capacitor are generated between the first external electrode 40A to which the first internal electrode layers 31 are connected and the second external electrode 40B to which the second internal electrode layers 32 are connected.

[0072] The first external electrode 40A includes a first base electrode layer 50A and a first plated layer 60A provided on the first base electrode layer 50A.

[0073] The second external electrode 40B includes a second base electrode layer 50B and a second plated layer 60B provided on the second base electrode layer 50B.

[0074] The first base electrode layer 50A is provided on the first end surface LS1. The first base electrode layer 50A is connected to the first internal electrode layer 31. In the present example embodiment of the present invention, the first base electrode layer 50A extends from the first end surface LS1 to a portion of the first main surface TS1 and to a portion of the second main surface TS2, and to a portion of the first lateral surface WS1 and to a portion of the second lateral surface WS2.

[0075] The second base electrode layer 50B is provided on the second end surface LS2. The second base electrode layer 50B is connected to the second internal electrode layer 32. In the present example embodiment of the present invention, the second base electrode layer 50B extends from the second end surface LS2 to a portion of the first main surface TS1 and to a portion of the second main surface TS2, and to a portion of the first lateral surface WS1 and to a portion of the second lateral surface WS2.

[0076] In the present example embodiment of the present invention, each of the first base electrode layer 50A and the second base electrode layer 50B is, for example, a fired layer. The fired layer preferably includes a metal component and either a glass component or a ceramic component, or alternatively, a metal component and both of a glass component and a ceramic component. The metal component includes, for example, at least one of Cu, Ni, Ag, Pd, Ag—Pd alloys, or Au. The glass component includes, for example, at least one of B, Si, Ba, Mg, Al, or Li. As the ceramic component, a ceramic material of the same kind as that of the dielectric layer 20 may be used, or a ceramic material of a different kind may be used. The ceramic component includes, for example, at least one of BaTiO₃, CaTiO₃, (Ba, Ca)TiO₃, SrTiO₃, or CaZrO₃.

[0077] The fired layer is obtained by applying a conductive paste including glass and metal to the multilayer body, and then firing. The fired layer may be obtained by simultaneously firing a multilayer (laminated) chip including the internal electrode layers and the dielectric layers, and an electrically conductive paste applied to the multilayer chip, or alternatively may be obtained by firing the multilayer chip including the internal electrode layers and the dielectric layers to thereby obtain a multilayer body, followed by the electrically conductive paste being applied to the multilayer body and then firing being performed. In a case in which the multilayer chip including the internal electrode layers and the dielectric layers, and the electrically conductive paste applied to the multilayer chip are fired simultaneously, it is preferable that the firing layer is formed by firing a material to which a ceramic material is added instead of the glass component. In this case, it is particularly preferable to use

the same type of ceramic material as the dielectric layer 20 as the ceramic material to be added. Furthermore, the fired layer may include a plurality of layers.

[0078] The thickness in the length direction of the first base electrode layer 50A located on the first end surface LS1 is preferably, for example, about 10 μm or more and about 200 μm or less at the middle portion in the height direction T and the width direction W of the first base electrode layer 50A.

[0079] The thickness in the length direction of the second base electrode layer 50B located on the second end surface LS2 is preferably, for example, about 10 μm or more and about 200 μm or less at the middle portion in the height direction T and the width direction W of the second base electrode layer 50B.

[0080] In a case in which the first base electrode layer 50A is provided on a portion of the surface of at least the first main surface TS1 or the second main surface TS2, it is preferable that the thickness in the height direction of the first base electrode layer 50A on the provided surface is, for example, about 5 μm or more and about 40 μm or less at the middle portion in the length direction L and the width direction W of the first base electrode layer 50A on the provided surface.

[0081] In a case in which the first base electrode layer 50A is provided on a portion of the surface of at least the first lateral surface WS1 or the second lateral surface WS2, it is preferable that the thickness in the width direction of the first base electrode layer 50A on the provided surface is, for example, about 5 μm or more and about 40 μm or less at the middle portion in the length direction L and the height direction T of the first base electrode layer 50A on the provided surface.

[0082] In a case in which the second base electrode layer 50B is provided on a portion of the surface of at least the first main surface TS1 or the second main surface TS2, it is preferable that the thickness in the height direction of the second base electrode layer 50B on the provided surface is, for example, about 5 μm or more and about 40 μm or less at the middle portion in the length direction L and the width direction W of the second base electrode layer 50B on the provided surface.

[0083] In a case in which the second base electrode layer 50B is provided on a portion of the surface of at least the first lateral surface WS1 or the second lateral surface WS2, it is preferable that the thickness in the width direction of the second base electrode layer 50B on the provided surface is, for example, about 5 μm or more and about 40 μm or less at the middle portion in the length direction L and the height direction T of the second base electrode layer 50B on the provided surface.

[0084] The first base electrode layer 50A and the second base electrode layer 50B are not limited to the fired layer, and each may be a thin film layer, for example. The thin film layer is a layer in which metal particles are deposited, and which is formed by a thin film forming method such as a sputtering method or a deposition method, for example. The thin film layer preferably includes, for example, at least one metal of Mg, Al, Ti, W, Cr, Cu, Ni, Ag, Co, Mo, or V. Thus, it is possible to increase the adhesion force of the external electrodes 40 to the multilayer body 10. The thin film layer may include a single layer or may include a plurality of layers. For example, the thin film layer may include a two-layer structure of a layer of NiCr and a layer of NiCu.

[0085] In a case in which the thin film layer defining and functioning as a base electrode is formed by a sputtered electrode by a sputtering method, for example, the sputtered electrode is preferably formed on a portion of the first main surface TS1 and on a portion of the second main surface TS2 of the multilayer body 10. The sputtered electrode preferably includes at least one metal of Ni, Cr, or Cu, for example. The thickness of the sputtered electrode is, for example, preferably about 50 nm or more and about 400 nm or less, and more preferably about 50 nm or more and about 130 nm or less.

[0086] As the base electrode layer, a sputtered electrode may be provided on a portion of the first main surface TS1 and on a portion of the second main surface TS2 of the multilayer body 10, while a fired layer may be provided on the first end surface LS1 and the second end surface LS2. Alternatively, the base electrode layer may not be provided on the first end surface LS1 and the second end surface LS2, and a plated layer, which will be described later, may be provided directly on the multilayer body 10. In addition, in a case in which a fired layer is provided on the first end surface LS1 and the second end surface LS2, the fired layer may be provided not only on the first end surface LS1 and the second end surface LS2, but also on a portion of the first main surface TS1 and on a portion of the second main surface TS2. In this case, the sputtered electrode may overlap the fired layer.

[0087] The first plated layer 60A covers the first base electrode layer 50A.

[0088] The second plated layer 60B covers the second base electrode layer 50B.

[0089] The first plated layer 60A and the second plated layer 60B may include at least one of Cu, Ni, Sn, Ag, Pd, Ag—Pd alloy, or Au, for example. Each of the first plated layer 60A and the second plated layer 60B may include a plurality of layers. The first plated layer 60A and the second plated layer 60B are preferably, for example, a two-layer structure in which a Sn-plated layer is provided on the Ni-plated layer.

[0090] The first plated layer 60A covers the first base electrode layer 50A. In the present example embodiment of the present invention, the first plated layer 60A includes, for example, a first Ni-plated layer 61A and a first Sn-plated layer 62A located on the first Ni-plated layer 61A.

[0091] The second plated layer 60B covers the second base electrode layer 50B. In the present example embodiment of the present invention, the second plated layer 60B includes, for example, a second Ni-plated layer 61B and a second Sn-plated layer 62B located on the second Ni-plated layer 61B.

[0092] The Ni-plated layer prevents the first base electrode layer 50A and the second base electrode layer 50B from being eroded by solder defining and functioning as the bonding material 5 for bonding the multilayer ceramic capacitor main body 2 and the metal terminal 100. Furthermore, the Sn-plated layer improves the wettability of the solder defining and functioning as the bonding material 5 (to be described later) for bonding the multilayer ceramic capacitor main body 2 and the metal terminal 100. This facilitates the bonding of the multilayer ceramic capacitor main body 2 and the metal terminal 100. In a case in which each of the first plated layer 70A and the second plated layer 70B is a two-layer structure of the Ni-plated layer and the Sn-plated layer, the thickness of each of the Ni-plated layer

and the Sn-plated layer is, for example, preferably about 1 μm or more and about 15 μm or less.

[0093] Furthermore, each of the first external electrode 40A and the second external electrode 40B of the present example embodiment may include an electrically conductive resin layer including, for example, electrically conductive particles and a thermosetting resin. In a case in which the electrically conductive resin layer is provided as the base electrode layer (the first base electrode layer 50A, the second base electrode layer 50B), the electrically conductive resin layer may cover the fired layer or may be provided directly on the multilayer body 10 without providing the fired layer. In a case in which the electrically conductive resin layer covers the fired layer, the conductive resin layer is provided between the fired layer and the plated layer (the first plated layer 60A, the second plated layer 60B). The electrically conductive resin layer may completely cover the fired layer or may partially cover the fired layer.

[0094] The electrically conductive resin layer including a thermosetting resin is more flexible than an electrically conductive layer made of, for example, a plating film or a fired product of an electrically conductive paste. Therefore, even when an impact caused by physical shock or thermal cycle to the multilayer ceramic capacitor 1 is applied, the electrically conductive resin layer functions as a buffer layer. Accordingly, crack generation of the multilayer ceramic capacitor 1 is reduced or prevented.

[0095] The metal of the electrically conductive particles may be, for example, Ag, Cu, Ni, Sn, Bi, or an alloy including them. The electrically conductive particles preferably include Ag, for example. The electrically conductive particles are metal powders of Ag, for example. Ag is suitable for electrode materials because of its lowest specific resistance among metals. Since Ag is a noble metal, it is difficult to spread and the weatherability is high. Therefore, the metal powder of Ag is suitable as electrically conductive particles.

[0096] Furthermore, the electrically conductive particles may be a metal powder in which the surface of the metal powder is coated with Ag, for example. In a case in which the metal powder coated with Ag is used, the metal powder is, for example, preferably Cu, Ni, Sn, Bi or an alloy powder thereof. In order to make the metal of the base material inexpensive while maintaining the Ag characteristics, for example, it is preferable to use a metal powder coated with Ag.

[0097] Furthermore, the electrically conductive particles may be provided by, for example, subjecting Cu or Ni to an oxidation prevention treatment. The electrically conductive particles may be a metal powder obtained by coating the surface of the metal powder with Sn, Ni, or Cu, for example. In a case in which the metal powder coated with Sn, Ni, or Cu is used, the metal powder is, for example, preferably Ag, Cu, Ni, Sn, or Bi or an alloy powder thereof.

[0098] The shape of the electrically conductive particles is not particularly limited. The electrically conductive particles may have a spherical shape, a flat shape, or the like. However, it is preferable to use a mixture of spherical and flat metal powders.

[0099] The electrically conductive particles included in the electrically conductive resin layer mainly provide the electric conductivity of the electrically conductive resin layer. More specifically, the plurality of electrically conduc-

tive particles are brought into contact with each other to provide an electric current-carrying path inside the electrically conductive resin layer.

[0100] The resin of the electrically conductive resin layer may include, for example, at least one of various known thermosetting resins such as epoxy resin, phenol resin, urethane resin, silicone resin, or polyimide resin. Among them, epoxy resins excellent in heat resistance, moisture resistance, adhesiveness and the like are suitable resins. Furthermore, the resin of the electrically conductive resin layer preferably includes a curing agent together with the thermosetting resin. In a case in which an epoxy resin is used as the base resin, the curing agent of the epoxy resin may be any of various known compounds such as, for example, phenolic, amine-based, acid anhydride-based, imidazole-based, active ester-based, or amideimide-based compounds.

[0101] In addition, the electrically conductive resin layer may include a plurality of layers. The thickness of the thickest portion of the electrically conductive resin layer is, for example, preferably about 10 μm or more and about 150 μm or less.

[0102] In addition, the first plated layer 60A and the second plated layer 60B (to be described later) may be directly provided on the multilayer body 10 without providing the first base electrode layer 50A and the second base electrode layer 50B. In other words, the multilayer ceramic capacitor 1 may include a plated layer that is electrically connected directly to the first internal electrode layers 31 and the second internal electrode layers 32. In such a case, a plated layer may be provided after placing a catalyst on the surface of the multilayer body 10 as a pretreatment.

[0103] Also in this case, the plated layer preferably includes a plurality of layers. Each of a lower plated layer and a lower plated layer preferably includes, for example, at least one metal of Cu, Ni, Sn, Pb, Au, Ag, Pd, Bi, Zn, or the like, or an alloy including these metals. The lower plated layer is, for example, more preferably made using Ni having solder barrier performance. The upper plated layer is, for example, more preferably made using Sn or Au having good solder wettability. Furthermore, in a case in which, for example, the first internal electrode layers 31 and the second internal electrode layers 32 are made using Ni, it is preferable that the lower plated layer is made using Cu having good bonding property with Ni. In addition, the upper plated layer may be provided as necessary, and the external electrode 40 may only include the lower plated layer. Furthermore, in the plated layer, the upper plated layer may be the outermost layer, or another plated layer may be further provided on the surface of the upper plated layer.

[0104] The thickness per layer of the plated layer without providing the base electrode layer is, for example, preferably about 2 μm or more and about 10 μm or less. The plated layer preferably does not include glass. The proportion of metal per unit volume of the plated layer is, for example, preferably about 99% by volume or more.

[0105] In a case in which the plated layer is provided directly on the multilayer body 10, it is possible to reduce the thickness of the base electrode layer. Therefore, it is possible to reduce the dimension in the height direction T of the multilayer ceramic capacitor main body 2 by the amount of the reduction in thickness of the base electrode layer, thus reducing the height of the multilayer ceramic capacitor main body 2. Alternatively, it is possible to increase the thickness of the dielectric layers 20 sandwiched between the first

internal electrode layers **31** and the second internal electrode layers **32** by the amount of the reduction in thickness of the base electrode layer, thus improving the thickness of the base body. In this way, by providing the plated layer directly on the multilayer body **10**, it is possible to improve the degree of freedom in designing the multilayer ceramic capacitor.

[0106] When the dimension in the length direction of the multilayer ceramic capacitor main body **2** including the multilayer body **10** and the external electrode **40** is defined as the dimension *L*, *L* is, for example, preferably about 0.2 mm or more and about 10 mm or less. When the dimension in the height direction of the multilayer ceramic capacitor main body **2** is defined as the dimension *T*, *T* is, for example, preferably about 0.1 mm or more and about 10 mm or less. When the dimension in the width direction of the multilayer ceramic capacitor main body **2** is defined as the dimension *W*, *W* is, for example, preferably about 0.1 mm or more and about 10 mm or less.

[0107] In the present example embodiment, the first surface **S1** on the first end surface **LS1** of the multilayer ceramic capacitor main body **2** is defined by the surface of the first external electrode **40A** provided on the first end surface **LS1**. The second surface **S2** on the second end surface **LS2** of the multilayer ceramic capacitor main body **2** is defined by the surface of the second external electrode **40B** provided on the second end surface **LS2**.

[0108] The metal terminal **100** will be described with reference to FIG. **11** in addition to FIGS. **1** to **6**. FIG. **11** is a view corresponding to FIG. **4**, and is an arrow view as seen in the height direction from the second main surface **TS2** toward the first main surface **TS1**, showing the metal terminal **100** when the exterior material **3** and the multilayer ceramic capacitor main body **2** are excluded. In FIG. **11**, the profile of the multilayer body **10** and the external electrode **40** of the multilayer ceramic capacitor main body **2** are indicated by a two-dot chain line.

[0109] The metal terminal **100** includes a first metal terminal **100A** and a second metal terminal **100B**.

[0110] The first metal terminal **100A** and the second metal terminal **100B** are metal terminals to be mounted on a mounting surface of a mounting substrate (refer to the mounting substrate **310** in FIGS. **15A** and **15B**) to be described later on which the multilayer ceramic capacitor **1** is to be mounted. The first metal terminal **100A** and the second metal terminal **100B** are, for example, plate-shaped lead frames. In an example embodiment of the present invention, the first main surface **TS1** of the multilayer body **10** is a surface opposed to the mounting surface of the mounting substrate on which the multilayer ceramic capacitor **1** is to be mounted.

[0111] The first metal terminal **100A** includes a first bonding portion **110A** that is opposed to the first main surface **TS1** and connected to the first external electrode **40A**, a first rising portion **120A** that is connected to the first bonding portion **110A**, extends away from the mounting surface of the mounting substrate, and is opposed to the first end surface **LS1**, a first extension portion **130A** that is connected to the first rising portion **120A** and extends away from the multilayer ceramic capacitor main body **2** in the length direction *L*, a first falling portion **140A** that is connected to the first extension portion **130A** and extends toward the mounting surface side of the mounting substrate, and a first mounting portion **150A** that is connected to the first falling

portion **140A** and extends in the direction along the mounting surface of the mounting substrate. As shown in FIGS. **6**, a gap portion *G* exists between the first rising portion **120A** and the first surface **S1** on the first end surface **LS1** of the multilayer ceramic capacitor main body **2**. Details of the first metal terminal **100A** will be described later.

[0112] The second metal terminal **100B** includes a second bonding portion **110B** that is opposed to the first main surface **TS1** and connected to the second external electrode **40B**, a second rising portion **120B** that is connected to the second bonding portion **110B**, extends away from the mounting surface of the mounting substrate, and is opposed to the second end surface **LS2**, a second extension portion **130B** that is connected to the second rising portion **120B** and extends away from the multilayer ceramic capacitor main body **2** in the length direction *L*, a second falling portion **140B** that is connected to the second extension portion **130B** and extends toward the mounting surface side of the mounting substrate, and a second mounting portion **150B** that is connected to the second falling portion **140B** and extends in the direction along the mounting surface of the mounting substrate. As shown in FIGS. **6**, a gap portion *G* exists between the second rising portion **120B** and the second surface **S2** on the second end surface **LS2** of the multilayer ceramic capacitor main body **2**. Details of the second metal terminal **100B** will be described later.

[0113] In addition, the first falling portion **140A** and the second falling portion **140B** preferably extend toward the mounting surface of the mounting substrate to an extent such that a gap can be provided between the exterior material **3** of the multilayer ceramic capacitor **1** and the mounting surface of the mounting substrate.

[0114] By providing such a first metal terminal **100A** and a second metal terminal **100B**, it is possible to lengthen the distance between the mounting substrate and the multilayer ceramic electronic component main body **2** such that it is possible to achieve an advantageous effect of relieving stress from the mounting substrate. Furthermore, the thickness of the exterior material **3** provided adjacent to the mounting border can be increased such that the insulating property can be ensured.

[0115] As shown in FIG. **11**, the separation distance *L4* between the first mounting portion **150A** of the first metal terminal **100A** and the second mounting portion **150B** of the second metal terminal **100B** is longer than the separation distance *L3* between the first external electrode **40A** and the second external electrode **40B** of the multilayer ceramic capacitor main body **2**.

[0116] The bonding material **5** joins the multilayer ceramic capacitor main body **2** and the metal terminal **100**. The bonding material **5** includes a first bonding material **5A** and a second bonding material **5B**.

[0117] As shown in FIG. **6**, the first metal terminal **100A** is connected to the first external electrode **40A** through the first bonding material **5A**. The second metal terminal **100B** is connected to the second external electrode **40B** via the second bonding material **5B**.

[0118] The bonding material **5** is preferably solder. For example, Pb-free solder may be used. As the Pb-free solder, for example, lead-free solder such as Sn—Sb solder, Sn—Ag—Cu solder, Sn—Cu solder, or Sn—Bi solder is preferable. For example, Sn-10Sb to Sn-15Sb solder can be used.

[0119] The exterior material **3** will be described with reference to FIGS. **1** to **6**.

[0120] The exterior material 3 includes a first main surface MTS1 and a second main surface MTS2 which are opposed to each other in the height direction T, a first lateral surface MWS1 and a second lateral surface MWS2 which are opposed to each other in the width direction W orthogonal or substantially orthogonal to the height direction T, and a first end surface MLS1 and a second end surface MLS2 which are opposed to each other in the length direction L orthogonal or substantially orthogonal to the height direction T and the width direction W. The first end surface MLS1 of the exterior material 3 is a surface of the exterior material 3 and is located adjacent to the first end surface LS1 of the multilayer body 10. The second end surface MLS2 of the exterior material 3 is a surface of the exterior material 3 and is located adjacent to the second end surface LS2 of the multilayer body 10.

[0121] The first lateral surface MWS1, the second lateral surface MWS2, the first end surface MLS1, and the second end surface MLS2 of the exterior material 3 include a parting line PL in the middle portion in the height direction T. The parting line PL is a line corresponding to a split surface of a mold for use in molding the exterior material 3. The surface of the exterior material 3 is provided with a draft angle with the parting line PL defining and functioning as a boundary.

[0122] The first lateral surface MWS1 of the exterior material 3 includes a first main surface-side surface MWS1A and a second main surface-side surface MWS1B. The second lateral surface MWS2 of the exterior material 3 includes a first main surface-side surface MWS2A and a second main surface-side surface MWS2B. The first end surface MLS1 of the exterior material 3 includes a first main surface-side surface MLS1A and a second main surface-side surface MLS1B. The second end surface MLS2 of the exterior material 3 includes a first main surface-side surface MLS2A and a second main surface-side surface MLS2B. The surface on the first main surface side and the surface on the second main surface side are separated from each other with the parting line PL as a boundary.

[0123] Each of the surfaces MWS1A, MWS2A, MLS1A and MLS2A on the first main surface side is provided with a draft angle such that the cross-sectional area of the cross section LW of the exterior material 3 becomes smaller as it approaches the first main surface TS1 from the parting line PL. Each of the surfaces MWS1B, MWS2B, MLS1B, and MLS2B on the second main surface side is provided with a draft angle such that the cross-sectional area of the cross section LW of the exterior material 3 becomes smaller as it approaches the second main surface TS2 from the parting line PL.

[0124] The exterior material 3 covers the multilayer ceramic capacitor main body 2, the bonding material 5 connecting the multilayer ceramic capacitor main body 2 and the metal terminal 100 with each other, and a portion of the metal terminal 100. More specifically, the exterior material 3 covers the entire multilayer ceramic capacitor main body 2, the entire first bonding material 5A and second bonding material 5B, a portion of the first metal terminal 100A, and a portion of the second metal terminal 100B.

[0125] For example, the exterior material 3 covers the entire first bonding portion 110A, the entire first rising portion 120A, and at least a portion of the first extension portion 130A of the first metal terminal 100A. Furthermore, the exterior material 3 covers the entire second bonding

portion 110B, the entire second rising portion 120B, and at least a portion of the second extension portion 130B of the second metal terminal 100B.

[0126] In the present example embodiment, the first extension portion 130A of the first metal terminal 100A protrudes from the first end surface MLS1 of the exterior material 3 and is partially exposed. The second extension portion 130B of the second metal terminal 100B protrudes from the second end surface MLS2 of the exterior material 3 and is partially exposed. More specifically, the first extension portion 130A of the first metal terminal 100A protrudes from the parting line PL of the first end surface MLS1 of the exterior material 3 and is partially exposed. The second extension portion 130B of the second metal terminal 100B protrudes from the parting line PL of the second end surface MLS2 of the exterior material 3 and is partially exposed.

[0127] The second main surface MTS2 of the exterior material 3 preferably has a planar shape with a predetermined flatness. With such a configuration, it is possible to prevent improper suction adhesion of a mounter of a mounting machine for use in mounting the multilayer ceramic capacitor 1 on the mounting substrate. Therefore, it is possible to reliably mount the multilayer ceramic capacitor 1 on the mounting substrate. As a result, it is possible to reduce or prevent the occurrence of mounting defects.

[0128] The minimum distance from the second main surface MTS2 of the exterior material 3 to the surface of the multilayer ceramic capacitor main body 2 is, for example, preferably about 100 μm or more and about 4000 μm or less. The minimum distance from the first main surface MTS1 of the exterior material 3 to the first bonding portion 110A of the first metal terminal 100A is, for example, preferably about 100 μm or more and about 4000 μm or less. The minimum distance from the first lateral surface MWS1 of the exterior material 3 to the surface of the multilayer ceramic capacitor main body 2 is, for example, preferably about 100 μm or more and about 4000 μm or less. The minimum distance from the second lateral surface MWS2 of the exterior material 3 to the surface of the multilayer ceramic capacitor main body 2 is, for example, preferably about 100 μm or more and about 4000 μm or less. The minimum distance from the first end surface MLS1 of the exterior material 3 to the surface of the multilayer ceramic capacitor main body 2 is, for example, preferably about 300 μm or more and about 5000 μm or less. The minimum distance from the second end surface MLS2 of the exterior material 3 to the surface of the multilayer ceramic capacitor main body 2 is, for example, preferably about 300 μm or more and about 5000 μm or less. The average distance in the length direction L from the surface MLS1A on the first main surface side of the first end surface MLS1 of the exterior material 3 to the first rising portion 120A of the first metal terminal 100A is, for example, preferably about 200 μm or more and about 4900 μm or less. The average distance in the length direction L from the surface MLS2A on the first main surface side of the second end surface MLS2 of the exterior material 3 to the second rising portion 120B of the second metal terminal 100B is, for example, preferably about 200 μm or more and about 4900 μm or less.

[0129] The exterior material 3 is, for example, preferably made of resin. For example, the exterior material 3 may be formed by molding engineering plastic by transfer molding, injection molding, or the like. In particular, for example, the material of the exterior material 3 preferably includes a

thermosetting epoxy resin. With such a configuration, adhesion between the exterior material **3**, and the multilayer ceramic capacitor main body **2** and the metal terminal **100** can be ensured, such that it is possible to achieve the advantageous effect of improving the withstand voltage and moisture resistance. The exterior material **3** may be formed, for example, by applying a liquid or powdery silicone-based or epoxy-based resin.

[0130] In this way, by the exterior material **3** covering the conductive metal portion such as the external electrode **40** and the metal terminal **100** over a wide range, it is possible to ensure the insulating surface distance (creeping distance) between the conductors. Furthermore, by covering the conductive metal portion over a wide range with the exterior material **3**, it is possible to avoid the risk of surface discharge.

[0131] The shape of the exterior material **3** is not particularly limited. For example, a truncated cone such as a truncated pyramid may be used. The shape of the corner portion of the exterior material **3** is not particularly limited, and may be rounded.

[0132] In addition to FIG. 5, FIGS. 6, and 11, with reference to FIGS. 12A to 12F, a description will be provided of a configuration around the bonding portion between the metal terminal **100** and the external electrode **40** of the multilayer ceramic capacitor main body **2**, and details of the metal terminal **100**.

[0133] FIG. 12A is an enlarged view of a portion XIIA of the multilayer ceramic capacitor **1** shown in FIG. 6, and is a view for explaining the configuration around the bonding portion between the first external electrode **40A** and the first metal terminal **100A**, and the details of the first metal terminal **100A**. FIG. 12B is an enlarged view of a portion XIIB of the multilayer ceramic capacitor **1** shown in FIG. 6, and is a view for explaining the configuration around the bonding portion between the second external electrode **40B** and the second metal terminal **100B**, and the details of the second metal terminal **100B**. FIG. 12C is a partial external perspective view of the first metal terminal **100A**.

[0134] As shown in FIG. 12A, a gap portion **G** exists between the first rising portion **120A** of the first metal terminal **100A** and the first surface **S1** on the first end surface **LS1** of the multilayer ceramic capacitor main body **2**, and the gap portion **G** is filled with the exterior material **3**. In the present example embodiment, the first surface **S1** is the surface of the first external electrode **40A** on the first end surface **LS1**. That is, in the present example embodiment, the gap portion **G** is provided between the first rising portion **120A** and the first surface **S1** of the first external electrode **40A** on the first end surface **LS1**, and the gap portion **G** is filled with the exterior material **3**. The average distance in the length direction **L** of the gap portion **G** is, for example, preferably about 50 μm or more and about 1500 μm or less. With such a configuration, it is possible to reliably prevent contact between the first external electrode **40A** and the first rising portion **120A** without increasing the dimensions of the multilayer ceramic capacitor **1**. In addition, it is possible to appropriately fill the gap portion **G** with the exterior material **3**, and it is possible to reduce or prevent the occurrence of solder splash or other problems during reflow at the time of substrate mounting.

[0135] The first rising portion **120A** is sloped away from the first surface **S1** on the first end surface **LS1** of the multilayer ceramic capacitor main body **2** from the connec-

tion portion with the first bonding portion **110A** toward the connection portion with the first extension portion **130A**. With such a configuration, the distance in the length direction **L** of the gap portion **G** increases from a position closer to the mounting surface of the mounting substrate as it approaches a position spaced farther away from the mounting surface. That is, a distance **G2** in the length direction **L** at a position spaced farther away from the mounting surface of the gap portion **G** is longer than a distance **G1** in the length direction **L** at a position closer to the mounting surface of the gap portion **G**. The angle α between the first rising portion **120A** and the first surface **S1** on the first end surface **LS1** of the multilayer ceramic capacitor main body **2** is, for example, preferably about 1° or more and about 40° or less.

[0136] Here, the surface **MLS1A**, which is a surface of the first end surface **MLS1** of the exterior material **3**, defines and functions as a first sloped surface of the exterior material **3**, and is located adjacent to the first main surface and closer to the mounting surface than the portion where the first extension portion **130A** protrudes. The first sloped surface **MLS1A** is sloped away from the first surface **S1** of the multilayer ceramic capacitor main body **2** from a position closer to the mounting surface as it approaches a position spaced farther away from the mounting surface. The draft angle β of the first sloped surface **MLS1A** is, for example, preferably about 1° or more and about 20° or less. The angle between the first rising portion **120A** and the first sloped surface **MLS1A** is, for example, preferably about 30° or less. In this way, the first rising portion **120A** and the first sloped surface **MLS1A** are sloped in the same or similar direction, and the difference between the slope angles of the first rising portion **120A** and the first sloped surface **MLS1A** is reduced, such that it is possible to make the distance from the first sloped surface **MLS1A** of the exterior material **3** to the first rising portion **120A** of the first metal terminal **100A** substantially constant. With such a configuration, it is possible to ensure the strength around the first rising portion **120A** to which a force is easily applied.

[0137] As shown in FIG. 12B, a gap portion **G** exists between the second rising portion **120B** of the second metal terminal **100B** and the second surface **S2** on the second end surface **LS2** of the multilayer ceramic capacitor main body **2**, and the gap portion **G** is filled with the exterior material **3**. In the present example embodiment, the second surface **S2** is the surface of the second external electrode **40B** on the second end surface **LS2**. That is, in the present example embodiment, the gap portion **G** is provided between the second rising portion **120B** and the second surface **S2** of the second external electrode **40B** provided on the second end surface **LS2**, and the gap portion **G** is filled with the exterior material **3**. The average distance in the length direction **L** of the gap portion **G** is, for example, preferably about 50 μm or more and about 1500 μm or less. With such a configuration, it is possible to reliably prevent the contact between the second external electrode **40B** and the second rising portion **120B** without increasing the dimensions of the multilayer ceramic capacitor **1**. In addition, it is possible to appropriately fill the gap portion **G** with the exterior material **3**, and it is possible to reduce or prevent the occurrence of solder splash or other problems during reflow at the time of substrate mounting.

[0138] The second rising portion **120B** is sloped away from the second surface **S2** on the second end surface **LS2**

of the multilayer ceramic capacitor main body **2** from the connection portion with the second bonding portion **110B** as it approaches the connection portion with the second extension portion **130B**. With such a configuration, the distance in the length direction **L** of the gap portion **G** increases from a position closer to the mounting surface of the mounting substrate as it approaches a position spaced farther away more from the mounting surface. That is, a distance **G2** in the length direction **L** at a position spaced farther away from the mounting surface of the gap portion **G** is longer than a distance **G1** in the length direction **L** at a position closer to the mounting surface of the gap portion **G**. The angle α between the second rising portion **120B** and the second surface **S2** on the second end surface **LS2** of the multilayer ceramic capacitor main body **2** is, for example, preferably about 1° or more and about 40° or less.

[0139] Here, the surface **MLS2A**, which is a surface of the second end surface **MLS2** of the exterior material **3**, defines and functions as a second sloped surface of the exterior material **3**, and is located adjacent to the first main surface side and closer to the mounting surface than the portion where the second extension portion **130B** protrudes. The second sloped surface **MLS2A** is sloped away from the second surface **S2** of the multilayer ceramic capacitor main body **2** from a position closer to the mounting surface toward a position spaced farther away from the mounting surface. The draft angle β of the second sloped surface **MLS2A** is, for example, preferably about 1° or more and about 20° or less. The angle between the second rising portion **120B** and the second sloped surface **MLS2A** is, for example, preferably about 30° or less. In this way, the second rising portion **120B** and the second sloped surface **MLS2A** are sloped in the same or similar direction, and the difference between the slope angles of the second rising portion **120B** and the second sloped surface **MLS2A** is reduced, such that it is possible to make the distance from the second sloped surface **MLS2A** of the exterior material **3** to the second rising portion **120B** of the second metal terminal **100B** substantially constant. With such a configuration, it is possible to ensure the strength around the second rising portion **120B** to which a force is easily applied.

[0140] The average distance in the length direction **L** from the surface **MLS1A** of the first main surface side of the first end surface **MLS1** of the exterior material **3** to the first rising portion **120A** of the first metal terminal **100A** is, for example, preferably about 0.133 times or more the average distance in the length direction **L** of the gap portion **G**. More preferably, for example, it is about 4 times or more and about 98 times or less. More preferably, for example, it is about 6 times or more and about 98 times or less. With such a configuration, it is possible to ensure the strength around the first rising portion **120A** to which a force is easily applied. It is also possible to improve moisture resistance.

[0141] The average distance in the length direction **L** from the surface **MLS2A** on the first main surface side of the second end surface **MLS2** of the exterior material **3** to the second rising portion **120B** of the second metal terminal **100B** is, for example, preferably about 0.133 times or more the average distance in the length direction **L** of the gap portion **G**. More preferably, for example, it is about 4 times or more and about 98 times or less. More preferably, for example, it is about 6 times or more and about 98 times or less. With such a configuration, it is possible to ensure the

strength around the second rising portion **120B** to which a force is easily applied. It is also possible to improve moisture resistance.

[0142] The measurement of the average distance in the length direction **L** of each of the measurement target portions such as the above-described gap portion **G** and a predetermined portion of the exterior material **3** is performed by the following method. First, the multilayer ceramic capacitor **1** is cross-sectionally polished to about one half in the **W** dimension to expose a specific **LT** cross section in which the cross section of the metal terminal **100** can be confirmed. Then, the **LT** cross section of the multilayer ceramic capacitor **1** exposed by polishing is observed by SEM. Next, in the measurement target portion, ten lines extending in the length direction **L** are drawn at equal or substantially equal intervals in the height **T** direction, and an average of distances of the ten lines is set as an average distance in the length direction **L** of the measurement target portion in the present example embodiment.

[0143] FIG. 12C is an external perspective view showing a portion of the appearance of the first metal terminal **100A** as an example of the metal terminal **100**. The first metal terminal **100A** and the second metal terminal **100B** are generally plane-symmetrical with respect to the cross section **WT** at the middle in the length direction **L** of the multilayer ceramic capacitor **1**. Therefore, the external perspective view (not shown) of the second metal terminal **100B** is the same or substantially the same as the external perspective view of the first metal terminal **100A**.

[0144] The first metal terminal **100A** includes a first notch **160A**, a first opening portion **170A**, and a third notch **180A**.

[0145] The first notch **160A** continuously extends from the end of the first bonding portion **110A** to a position in the middle of the first rising portion **120A**. With such a configuration, for example, when the exterior material **3** is molded, the resin of the exterior material **3** flows through the first notch **160A**, such that the gap portion **G** is easily filled with the resin. Furthermore, since the resin of the exterior material **3** is provided in the first notch **160A**, the resin on one surface side and the resin on the other surface side of the first rising portion **120A** of the first metal terminal **100A** are connected by the resin in the first notch **160A**, such that the structure becomes stronger. Since the cut-away portion of the first notch **160A** extends a position in the middle of the first rising portion **120A**, the strength of the first metal terminal **100A** is ensured. Since the first rising portion **120A** of the present example embodiment is sloped as described above, for example, during molding of the exterior material **3**, the resin of the exterior material **3** is likely to enter the gap portion **G** and flow through the first notch **160A**.

[0146] As shown in FIG. 12C, the rising height **T3** of the first notch **160A** in the height direction **T** is, for example, preferably about half or less the rising height **T2** of the first rising portion **120A** in the height direction **T**. With such a configuration, for example, when the exterior material **3** is molded, it is possible to ensure the strength of the first metal terminal **100A**, while ensuring the flowability of the resin of the exterior material **3**.

[0147] The first bonding portion **110A** includes a first bonding piece **111A** adjacent to the first lateral surface **WS1** and a second bonding piece **112A** adjacent to the second lateral surface **WS2** which are divided by the first notch **160A**.

[0148] The first opening portion 170A is provided at the first extension portion 130A. As described above, by providing the first opening portion 170A in addition to the first notch 160A in the first metal terminal 100A, it is possible to further improve the flowability of the resin of the exterior material 3 during molding of the exterior material 3, for example. Furthermore, since the resin of the exterior material 3 is provided in the first opening portion 170A, the resin on one surface side and the resin on the other surface side of the first extension portion 130A of the first metal terminal 100A are connected by the resin provided in the first opening portion 170A, such that the structure becomes stronger. According to the above configuration, it is preferable that the same material of the exterior material 3 is provided in the portion of the first notch 160A provided in the first rising portion 120A and the first opening portion 170A. With such a configuration, the structure of the multilayer ceramic capacitor 1 becomes stronger.

[0149] The third notch 180A continuously extends from the end of the first mounting portion 150A to a position in the middle of the first falling portion 140A.

[0150] As shown in FIGS. 11 and 12C, the length W2 in the width direction of the first bonding portion 110A of the first metal terminal 100A is longer than the length W3 in the width direction of the first rising portion 120A. With such a configuration, it is possible to ensure a wide bonding area between the first external electrode 40A and the first metal terminal 100A by the first bonding material 5A. In particular, even when the first notch 160A is provided as described above, it is still possible to ensure a wide bonding area between the first external electrode 40A and the first metal terminal 100A by the first bonding material 5A.

[0151] The length W4 in the width direction W of the first notch 160A may be equal to or substantially equal to the length W5 in the width direction W of the first opening portion 170A. The rising height T3 of the first notch 160A in the height direction T may be the same or approximately the same as the length L6 in the length direction L of the first opening portion 170A. For example, the area of the first notch 160A in the first rising portion 120A may fall within a range from about 50% to about 200% of the area of the first opening portion 170A. With such a configuration, for example, when the exterior material 3 is molded, the resin of the exterior material 3 flows in a well-balanced manner.

[0152] The second metal terminal 100B includes a second notch 160B, a second opening portion 170B, and a fourth notch 180B.

[0153] The second notch 160B continuously extends from the end of the second bonding portion 110B to a position in the middle of the second rising portion 120B. With such a configuration, for example, when the exterior material 3 is molded, the resin of the exterior material 3 flows through the second notch 160B, such that the gap portion G is easily filled with the resin. Furthermore, since the resin of the exterior material 3 is provided in the second notch 160B, the resin on one surface side and the resin on the other surface side of the second rising portion 120B of the second metal terminal 100B are connected by the resin in the second notch 160B, such that the structure becomes stronger. Since the cut-away portion of the second notch 160B extends to a position in the middle of the second rising portion 120B, the strength of the second metal terminal 100B is ensured. Since the second rising portion 120B of the present example is sloped as described above, for example, during molding of

the exterior material 3, the resin of the exterior material 3 is likely to enter the gap portion G and flow through the second notch 160B.

[0154] The rising height T3 of the second notch 160B in the height direction T is, for example, preferably about half or less the rising height T2 of the second rising portion 120B in the height direction T. With such a configuration, for example, when the exterior material 3 is molded, it is possible to ensure the strength of the second metal terminal 100B while ensuring the flowability of the resin of the exterior material 3.

[0155] The second bonding portion 110B includes a third bonding piece 111B adjacent to the first lateral surface WS1 and a fourth bonding piece 112B adjacent to the second lateral surface WS2 which are divided by the second notch 160B.

[0156] The second opening portion 170B is provided at the second extension portion 130B. With such a configuration, by providing the second metal terminal 100B with the second opening portion 170B in addition to the second notch 160B described above, it is possible to further improve the flowability of the resin of the exterior material 3 during molding of the exterior material 3, for example. Furthermore, since the resin of the exterior material 3 is provided in the second opening portion 170B, the resin on one surface side and the resin on the other surface side of the second extension portion 130B of the second metal terminal 100B are connected by the resin in the second opening portion 170B, such that the structure becomes stronger. According to the above configuration, it is preferable that the same material of the exterior material 3 is provided in the portion of the second notch 160B provided in the second rising portion 120B and the second opening portion 170B. With such a configuration, the structure of the multilayer ceramic capacitor 1 becomes stronger.

[0157] The fourth notch 180B continuously extends from the end of the second mounting portion 150B to a position in the middle of the second falling portion 140B.

[0158] As shown in FIG. 11, the length W2 in the width direction of the second bonding portion 110B of the second metal terminal 100B is longer than the length W3 in the width direction of the second rising portion 120B. With such a configuration, it is possible to ensure a wide bonding area between the second external electrode 40B and the second metal terminal 100B by the second bonding material 5B. In particular, even when the second notch 160B is provided as described above, it is still possible to ensure a wide bonding area between the second external electrode 40B and the second metal terminal 100B by the second bonding material 5B.

[0159] The length W4 in the width direction W of the second notch 160B may be equal to or substantially equal to the length W5 in the width direction W of the second opening portion 170B. The rising height T3 of the second notch 160B in the height direction T may be the same or approximately the same as the length L6 in the length direction L of the second opening portion 170B. For example, the area of the second notch 160B provided in the second rising portion 120B may fall within a range from about 50% to about 200% of the area of the second opening portion 170B. With such a configuration, for example, when the exterior material 3 is molded, the resin of the exterior material 3 flows in a well-balanced manner.

[0160] The first mounting portion **150A** may extend parallel or substantially parallel to the mounting surface along the mounting surface, or may extend to be sloped in a direction away from the mounting surface as it approaches the connection portion with the first falling portion **140A**. The second mounting portion **150B** may extend parallel or substantially parallel to the mounting surface along the mounting surface, or may extend to be sloped in a direction away from the mounting surface as it approaches the connection portion with the second falling portion **140B**. With such a configuration, when the multilayer ceramic capacitor **1** is mounted on the mounting substrate, it is possible for the bonding material to extend to this portion, and thus, it is possible to increase the strength in mounting. The slope angle θ is, for example, preferably about 1° or more and about 10° or less.

[0161] Next, with reference to FIGS. **12D** to **12F** in addition to FIGS. **12A** and **12B**, the state of the surface of the metal terminal **100** and the like will be described as further details of the metal terminal **100**.

[0162] FIG. **12D** is an enlarged view of a portion **R1** of the first metal terminal **100A** of the multilayer ceramic capacitor **1** shown in FIG. **12A**. FIG. **12E** is an enlarged view of a portion **R2** of the first metal terminal **100A** of the multilayer ceramic capacitor **1** shown in FIG. **12A**. FIG. **12F** is an enlarged view of a portion **R3** of the first metal terminal **100A** of the multilayer ceramic capacitor **1** shown in FIG. **12A**. As described above, the first metal terminal **100A** and the second metal terminal **100B** are substantially plane-symmetric with respect to the WT cross section at the middle in the length direction **L** of the multilayer ceramic capacitor **1**. Therefore, the enlarged view of the second metal terminal **100B** has the same or substantially the same shape as the enlarged view of the first metal terminal **100A**, which is left/right symmetrical with respect to the plane of the drawing. Therefore, in FIGS. **12D** to **12F**, in addition to the reference numerals denoting the respective configurations of the first metal terminal **100A**, the reference numerals in the second metal terminal **100B** are also denoted, and FIGS. **12D** to **12F** are used as enlarged views for explaining the first metal terminal **100A** and the second metal terminal **100B**.

[0163] As shown in FIG. **12A**, the first metal terminal **100A** is a plate-shaped member including a first front surface **FS1** adjacent to a first bonding surface **110A1** to which the first external electrode **40A** is bonded, a first opposite surface **BS1** which is a surface opposite to the first front surface **FS1**, and a first terminal lateral surface **TSS1** connecting the first front surface **FS1** and the first opposite surface **BS1**. The first bonding portion **110A** of the first metal terminal **100A** includes the first bonding surface **110A1** bonded to the first bonding material **5A** on the first front surface **FS1**.

[0164] As shown in FIG. **12B**, the second metal terminal **100B** is a plate-shaped member including a second front surface **FS2** on a second bonding surface **110B1** to which the second external electrode **40B** is bonded, a second opposite surface **BS2** which is a surface opposite to the second front surface **FS2**, and a second terminal lateral surface **TSS2** connecting the second front surface **FS2** and the second opposite surface **BS2**. The second bonding portion **110B** of the second metal terminal **100B** includes a second bonding surface **110B1** bonded to the second bonding material **5B** on the second front surface **FS2**.

[0165] As shown in FIGS. **12D** to **12F**, the first metal terminal **100A** includes a first base material **100Aa** defining a terminal main body and a first plating film **100Ab** provided on a surface of the terminal main body.

[0166] The first plating film **100Ab** of the first metal terminal **100A** is provided at least at a portion of the first bonding portion **110A** where the first bonding material **5A** is provided and at a portion of the first mounting portion **150A** that is opposed to the mounting surface of the mounting substrate.

[0167] As shown in FIGS. **12D** to **12F**, the second metal terminal **100B** includes a second base material **100Ba** defining a terminal main body and a second plating film **100Bb** provided on a surface of the terminal main body.

[0168] The second plating film **100Bb** of the second metal terminal **100B** is provided at least at a portion of the second bonding portion **110B** where the second bonding material **5B** is provided and at a portion of the second mounting portion **150B** that is opposed to the mounting surface of the mounting substrate.

[0169] The plating film preferably includes an upper plating film provided on the outermost surface of the plating film and a lower plating film provided below the upper plating film. For example, the plating film may include a two-layer structure in which an upper plating film is provided on a lower plating film.

[0170] In the first metal terminal **100A** according to the present example embodiment, the first plating film **100Ab** includes a first base plating film **100Ab1** defining and functioning as a lower plating film covering the surface of the first base material **100Aa** and a first outermost surface plating film **100Ab2** defining and functioning as an upper plating film covering the surface of the first base plating film **100Ab1**. In other words, the first outermost surface plating film **100Ab2** defining and functioning as the first outermost surface metal film includes at least the outermost surface portion of the first plating film **100Ab**.

[0171] Further, in the second metal terminal **100B** according to the present example embodiment, the second plating film **100Bb** includes a second base plating film **100Bb1** defining and functioning as a lower plating film covering the surface of the second base material **100Ba** and a second outermost surface plating film **100Bb2** defining and functioning as an upper plating film covering the surface of the second base plating film **100Bb1**. In other words, the second outermost surface plating film **100Bb2** defining and functioning as the second outermost surface metal film includes at least the outermost surface portion of the second plating film **100Bb**.

[0172] In the two-layer configuration of the plating films, the upper layer plating film provided on the outermost surface of the plating film has a surface higher in solder wettability than the surface of the metal of the base material of the terminal body. Further, in the two-layer configuration, the upper plating film provided on the outermost surface of the plating film has a surface higher in solder wettability than the surface of the lower plating film.

[0173] The lower plating film is, for example, preferably made of Ni, Fe, Cu, Ag, Cr, or an alloy including at least one of these metals as a main component. More preferably, for example, the lower plating film is made of Ni, Fe, Cr, or an alloy including at least one of these metals as a main component.

[0174] When the lower plating film is made of Ni, Fe, Cr, or an alloy including at least one of these metals as a main component, the heat resistance of the metal end can be improved.

[0175] In the first metal terminal 100A according to the present example embodiment, the first base plating film 100Ab1 defining and functioning as the lower layer plating film is, for example, a Ni plating film. In addition, the thickness of the first base plating film 100Ab1 as the lower plating film is, for example, preferably about 0.2 μm or more and about 5.0 μm or less. In the second metal terminal 100B according to the present example embodiment, the second base plating film 100Bb1 defining and functioning as the lower layer plating film is, for example, a Ni plating film. In addition, the thickness of the second base plating film 100Bb1 as the lower plating film is, for example, preferably about 0.2 μm or more and about 5.0 μm or less.

[0176] The upper plating film is, for example, preferably made of Sn, Ag, Au, or an alloy including at least one of these metals as a main component. More preferably, for example, the upper plating film is made of Sn or an alloy including Sn as a main component. By forming the upper plating film with Sn or an alloy including Sn as a main component, the solderability between the external electrode and the metal terminal can be improved.

[0177] In the first metal terminal 100A according to the present example embodiment, the first outermost surface plating film 100Ab2 defining and functioning as the upper layer plated layer is, for example, a Sn plating film. The thickness of the second base plating film 100Bb1 as the upper plating film is, for example, preferably about 1.0 μm or more and about 5.0 μm or less. In the second metal terminal 100B according to the present example embodiment, the second outermost surface plating film 100Bb2 defining and functioning as the upper layer plated layer is, for example, a Sn plating film. The thickness of the second base plating film 100Bb1 as the upper plating film is, for example, preferably about 1.0 μm or more and about 5.0 μm or less.

[0178] The terminal body is, for example, preferably made of Ni, Fe, Cu, Ag, Cr, or an alloy including at least one of these metals as a main component. Examples of the metal of the base material of the terminal body include an Fe-42Ni alloy, an Fe-18Cr alloy, or a Cu-8Sn alloy. In addition, for example, from the viewpoint of heat dissipation, the metal of the base material of the terminal main body may be oxygen-free copper or a Cu-based alloy having high thermal conductivity. By making the material of the terminal main body a copper-based material having good thermal conductivity in this way, it is possible to achieve a reduction in ESR and a reduction in thermal resistance. Further, in the present example embodiment, the metal of the base material of the terminal main body may be, for example, stainless steel or aluminum which have low solder wettability. At least the surface of the metal of the base material of the terminal main body is a surface having lower wettability of solder than the plating film of the outermost surface. The thickness of the terminal body is, for example, preferably about 0.05 mm or more and about 0.5 mm or less.

[0179] Further, the first metal terminal 100A includes a first contact surface CS1 in contact with the exterior material 3. The first contact surface CS1 of the first metal terminal 100A according to the present example embodiment includes, as surfaces in contact with the exterior material 3,

the first base material 100Aa, a surface of the first base plating film 100Ab1, and a surface of the first outermost surface plating film 100Ab2.

[0180] The first metal terminal 100A includes exposed surfaces E1a, E1b, and E1c where a material located on the inner side of the first outermost surface plating film 100Ab2 of the first plating film 100Ab is exposed to the surface in a portion of the surface in contact with the exterior material 3. FIG. 12A shows an example of the arrangement position of the exposed surfaces of the first metal terminal 100A.

[0181] The exposed surfaces E1a, E1b, and E1c may be surfaces on which the lower first base plating film 100Ab1 is exposed. The first base plating film 100Ab1 has lower solder wettability than the first outermost surface plating film 100Ab2 functioning as the upper layer. Therefore, it is possible to prevent the first bonding material 5A from further spreading along the first rising portion 120A of the first metal terminal 100A, and it is possible to further reduce or prevent the occurrence of solder splash or other problems.

[0182] The exposed surfaces E1a, E1b, and E1c may be surfaces to which the first base material 100Aa of the terminal main body is exposed. The solder wettability of the surface of the first base material 100Aa of the terminal main body is lower than that of the first outermost surface plating film 100Ab2 defining and functioning as the upper layer. Therefore, it is possible to prevent the first bonding material 5A from further spreading along the first rising portion 120A of the first metal terminal 100A, and it is possible to further reduce or prevent the occurrence of solder splash or other problems.

[0183] In the present example embodiment, the first contact surface CS1 in contact with the exterior material 3 includes a plurality of first exposed surfaces where the first base plating film 100Ab1 defining and functioning as a surface of a metal different from the first outermost surface plating film 100Ab2 functioning as the first outermost surface metal film is exposed. As the first exposed surface, for example, the exposed surface E1a, the exposed surface E1b, and the exposed surface E1c are provided.

[0184] The first contact surface CS1 in contact with the exterior material 3 includes the first outermost surface plating film 100Ab2 defining and functioning as a first outermost surface metal film on at least a portion of the surface.

[0185] The exposed surfaces E1b and E1a, each defining and functioning as the first exposed surfaces provided on the first contact surface CS1, are provided to be spaced apart on at least a portion of the surface between the middle of the first rising portion 120A and the first bonding portion 110A, and the first extension portion 130A, respectively, on the first front surface FS1. The exposed surfaces E1c and E1a, each defining and functioning as the first exposed surfaces provided on the first contact surface CS1, are provided to be spaced apart respectively on the first bonding portion 110A and the first extension portion 130A, on the first opposite surface BS1.

[0186] More specifically, as shown in FIG. 12D, the exposed surface E1a defining and functioning as the first exposed surface provided on the first contact surface CS1 is provided on the first front surface FS1 and the first opposite surface BS1 of the first extension portion 130A. The exposed surface E1a is provided on the first extension portion 130A adjacent to the first rising portion 120A. In the present example embodiment, the exposed surface E1a is

covered with the exterior material 3. That is, the exposed surface E1a is not exposed from the exterior material 3.

[0187] In addition, on the first extension portion 130A, the first contact surface CS1 includes the first base plating film 100Ab1 and the first outermost surface plating film 100Ab2 located on the first front surface FS1, the first base plating film 100Ab1 and the first outermost surface plating film 100Ab2 located on the first opposite surface BS1, and a surface of the first base material 100Aa located on the first terminal lateral surface TSS1. As described above, the exposed surface is preferably provided over the entire or substantially the entire circumference of the first metal terminal 100A in a portion of the extending direction of the first metal terminal 100A. Thus, the first outermost surface plating film 100Ab2 is divided in the middle in the extending direction of the first metal terminal 100A.

[0188] As shown in FIG. 12E, the exposed surface E1b defining and functioning as the first exposed surface provided on the first contact surface CS1 is provided on the first rising portion 120A of the first metal terminal 100A on the first front surface FS1. For example, the exposed surface E1b is provided to the first rising portion 120A adjacent to the connection-portion with the first bonding portion 110A.

[0189] In addition, the first contact surface CS1 includes the first base plating film 100Ab1 located on the first front surface FS1, the first outermost surface plating film 100Ab2 located on the first opposite surface BS1, and the surface of the first base material 100Aa located on the first terminal lateral surface TSS1, in at least a portion of the surface between the middle of the first rising portion 120A and the first bonding portion 110A.

[0190] Further, as shown in FIG. 12F, the exposed surface E1c defining and functioning as a first exposed surface provided on the first contact surface CS1 is provided on the first opposite surface BS1 of the first bonding portion 110A. For example, the exposed surface E1c is provided on the first contact surface CS1 of the first opposite surface BS1 of the first bonding portion 110A.

[0191] In addition, the first contact surface CS1 includes the first outermost surface plating film 100Ab2 located on the first front surface FS1, the first base plating film 100Ab1 located on the first opposite surface BS1, and a surface of the first base material 100Aa located on the first terminal lateral surface TSS1, in the first bonding portion 110A.

[0192] In addition, for example, since the first base plating film 100Ab1 is a Ni plating film, the exposed surfaces E1a, E1b, and E1c defining and functioning as the first exposed surfaces are Ni plating films of the first metal terminal 100A.

[0193] The exposed surfaces E1a, E1b, and E1c are not limited thereto. For example, on the exposed surfaces E1a, E1b, and E1c, the first base material 100Aa may be exposed without providing the first plating film 100Ab on the first base material 100Aa. In other words, the exposed surfaces E1a, E1b, and E1c defining and functioning as the first exposed surfaces may be the first base material 100Aa of the first metal terminal 100A.

[0194] In addition, the exposed surfaces E1a, E1b, and E1c defining and functioning as the first exposed surfaces provided on the first contact surface CS1 are separated in the width direction by holes or notches provided in the first metal terminal 100A.

[0195] Further, the second metal terminal 100B includes a second contact surface CS2 in contact with the exterior material 3. The second contact surface CS2 of the second

metal terminal 100B according to the present example embodiment includes, as surfaces in contact with the exterior material 3, the second base material 100Ba, the surface of the second base plating film 100Bb1, and the surface of the second outermost surface plating film 100Bb2.

[0196] The second metal terminal 100B includes exposed surfaces E2a, E2b, and E2c where the material located on the inner side of the second outermost surface plating film 100Bb2 of the second plating film 100Bb is exposed to the surface, in a portion of the surface in contact with the exterior material 3. FIG. 12B shows an example of the arrangement position of the exposed surfaces of the second metal terminal 100B.

[0197] The exposed surfaces E2a, E2b, and E2c may be surfaces where the lower second base plating film 100Bb1 is exposed. The solder wettability of the second base plating film 100Bb1 is lower than that of the second outermost surface plating film 100Bb2 defining and functioning as the upper layer. Therefore, it is possible to prevent the second bonding material 5B from further spreading along the second rising portion 120B of the second metal terminal 100B, and it is possible to further reduce or prevent the occurrence of solder splash or other problems.

[0198] The exposed surfaces E2a, E2b, and E2c may be surfaces to which the second base material 100Ba of the terminal main body is exposed. The solder wettability of the surface of the second base material 100Ba of the terminal main body is lower than that of the second outermost surface plating film 100Bb2 defining and functioning as the upper layer. Therefore, it is possible to prevent the second bonding material 5B from further spreading along the second rising portion 120B of the second metal terminal 100B, and it is possible to further reduce or prevent the occurrence of solder splash or other problems.

[0199] In the present example embodiment, the second contact surface CS2 in contact with the exterior material 3 includes a plurality of second exposed surfaces where the second base plating film 100Bb1 defining and functioning as a surface of a metal different from the second outermost surface plating film 100Bb2 defining and functioning as the second outermost surface metal film is exposed. As the second exposed surface, for example, the exposed surface E2a, the exposed surface E2b, and the exposed surface E2c are provided.

[0200] The second contact surface CS2 in contact with the exterior material 3 includes the second outermost surface plating film 100Bb2 defining and functioning as a second outermost surface metal film on at least a portion of the surface.

[0201] The exposed surfaces E2b and E2a each defining and functioning as the second exposed surfaces provided on the second contact surface CS2 are provided to be spaced apart on at least a portion of the surface between the middle of the second rising portion 120B and the second bonding portion 110B, and the second extension portion 130B, respectively, on the second front surface FS2. The exposed surfaces E2c and E2a each defining and functioning as the second exposed surfaces provided on the second contact surface CS2 are provided to be spaced apart respectively on the second bonding portion 110B and the second extension portion 130B, on the second opposite surface BS2.

[0202] More specifically, as shown in FIG. 12D, the exposed surface E2a defining and functioning as the second exposed surface provided on the second contact surface CS2

is provided on the second front surface FS2 and the second opposite surface BS2 of the second extension portion 130B. The exposed surface E2a is provided on the second extension portion 130B adjacent to the second rising portion 120B. In the present example embodiment, the exposed surface E2a is covered with the exterior material 3. That is, the exposed surface E2a is not exposed from the exterior material 3.

[0203] In addition, on the second extension portion 130B, the second contact surface CS2 includes the second base plating film 100Bb1 and the second outermost surface plating film 100Bb2 located on the second front surface FS2, the second base plating film 100Bb1 and a second outermost surface plating film 100Bb2 located on the second opposite surface BS2, and a surface of the second base material 100Ba located on the second terminal lateral surface TSS2. As described above, the exposed surface is preferably provided over the entire or substantially the entire circumference of the second metal terminal 100B in a portion in the extending direction of the second metal terminal 100B. Thus, the second outermost surface plating film 100Bb2 is divided in the middle in the extending direction of the second metal terminal 100B.

[0204] As shown in FIG. 12E, an exposed surface E2b defining and functioning as the second exposed surface provided on the second contact surface CS2 is provided on the second rising portion 120B of the second metal terminal 100B on the second front surface FS2. For example, the exposed surface E2b is provided to the second rising portion 120B adjacent to the connection-portion with the second bonding portion 110B.

[0205] In addition, the second contact surface CS2 includes the second base plating film 100Bb1 located on the second front surface FS2, the second outermost surface plating film 100Bb2 located on the second opposite surface BS2, and the surface of the second base material 100Ba located on the second terminal lateral surface TSS2, in at least a portion of the surface between the middle of the second rising portion 120B and the second bonding portion 110B.

[0206] Further, as shown in FIG. 12F, an exposed surface E2c defining and functioning as a second exposed surface provided on the second contact surface CS2 is provided on the second opposite surface BS2 of the second bonding portion 110B. For example, the exposed surface E2c is provided on the second contact surface CS2 of the second opposite surface BS2 of the second bonding portion 110B.

[0207] In addition, the second contact surface CS2 includes the second outermost surface plating film 100Bb2 located on the second front surface FS2, the second base plating film 100Bb1 located on the second opposite surface BS2, and a surface of the second base material 100Ba located on the second terminal lateral surface TSS2, in the second bonding portion 110B.

[0208] In addition, for example, since the second base plating film 100Bb1 is a Ni plating film, the exposed surfaces E2a, E2b, and E2c defining and functioning as the second exposed surfaces are Ni plating films of the second metal terminal 100B.

[0209] The exposed surfaces E2a, E2b, and E2c are not limited thereto. For example, on the exposed surfaces E2a, E2b, and E2c, the second base material 100Ba may be exposed without providing the second plating film 100Bb on the second base material 100Ba. In other words, the exposed

surfaces E2a, E2b, and E2c defining and functioning as the second exposed surfaces may be the second base material 100Ba of the second metal terminal 100B.

[0210] In addition, the exposed surfaces E2a, E2b, and E2c defining and functioning as the second exposed surfaces provided on the second contact surface CS2 are separated in the width direction by holes or notches provided in the second metal terminal 100B.

[0211] As shown in FIG. 13, a plurality of protruding portions U are provided on the first exposed surfaces E1a to E1c and the second exposed surfaces E2a to E2c, and the plurality of protruding portions U each have an undercut shape. Here, in the present specification, the undercut shape refers to a protruding portion shape including a space which is hidden by a portion of the protruding portion provided on a surface when the surface (for example, the contact surface of the metal terminal) is viewed in a direction orthogonal or substantially orthogonal to the surface.

[0212] In other words, in the undercut shape, when the exposed surface is viewed in the direction orthogonal or substantially orthogonal to the exposed surface, a portion of the rising shape portion of the protruding portion U is not seen by the protruding portion U itself. For example, the protruding portion U may be provided so that a portion of the surface of the first exposed surface E1a, E1b, or E1c or the surface of the second exposed surface E2a, E2b, or E2c, when viewed in a direction orthogonal or substantially orthogonal to the surface, hides and prevents another portion of the protruding portion U from being viewed. An example of the undercut shape of each of the protruding portions U according to the present example embodiment will be described below with reference to FIG. 13. FIG. 13 is a cross-sectional view showing an example of a plurality of protruding portions provided on an exposed surface of the first metal terminal. Since the plurality of protruding portions provided on the exposed surface of the second metal terminal are the same or substantially the same as those of the first metal terminal, descriptions thereof will be omitted.

[0213] In FIG. 13, the vertical direction in the drawing corresponds to a direction orthogonal or substantially orthogonal to the surface of the exposed surface. A virtual line V1 in FIG. 13 is parallel or substantially parallel to the vertical direction of the drawings and passes through a boundary point between adjacent protruding portions U. A virtual line V2 in FIG. 13 is parallel or substantially parallel to the vertical direction of the drawings and passes through the apex of each of the protruding portions U.

[0214] In the present example embodiment, as shown in FIG. 13, the undercut shape of each of the protruding portions U is configured so as to include, in its cross-sectional shape, a first side u1 parallel or substantially parallel to the surface (for example, the first exposed surface and the second exposed surface of the metal terminal) and defining and functioning as a reference line, a second side u2 extending from the first side u1 at a rising angle exceeding a right angle, and a third side u3 extending from the second side u2 so as to be connected to the first side u1. That is, the cross-sectional shape of each of the protruding portions U is a substantially obtuse triangle in which the angle θ between the first side u1 defining and functioning as the reference line of the surface and the third side u3 forming one of the rising portions of the protruding portion U is an obtuse angle. The term “substantially obtuse triangle” as used herein does not need to be an exact obtuse triangle, and includes, for

example, a shape in which the second side $u2$ has a skirt shape that gradually becomes an angle closer to parallel to the first side $u1$ as approaching the first side $u1$ as a reference line.

[0215] Therefore, in the example shown in FIG. 13, when the exposed surface is viewed from the direction orthogonal or substantially orthogonal to the exposed surface, in each of the protruding portions U on the right side in the drawings among the adjacent protruding portions U, a portion thereof located on the left side of the virtual line V2 is blocked by a portion of each of the protruding portions U on the left side in the drawings among the adjacent protruding portions U, and cannot be seen.

[0216] In another point of view, in the example shown in FIG. 13, when the exposed surface is viewed from the direction orthogonal or substantially orthogonal to the exposed surface, a portion H2 surrounded by the virtual lines V1 and V2 in each of the protruding portions U on the right side in the drawing among the adjacent protruding portions U may be hidden by a portion H1 surrounded by the virtual lines V1 and V2 in each of the protruding portions U on the left side in the drawings among the adjacent protruding portions U, and may not be seen.

[0217] With such a configuration, an anchor effect is generated between the exterior material and the plurality of protruding portions provided on the exposed surface of the metal terminal, and the adhesion between the metal terminal and the exterior material is increased. Specifically, the mold resin of the exterior material enters the space behind the protruding portion in the undercut shape, so that a high anchor effect is obtained.

[0218] Further, a plurality of protruding portions are continuously provided in the exposed surface of the metal terminal. Preferably, at least a portion of the plurality of protruding portions includes a large number of regularly arranged protruding portions. For example, the plurality of protruding portions are regularly arranged for each processed line.

[0219] The arithmetic mean height Sa of the exposed surface (the first exposed surfaces E1a, E1b, and E1c, and the second exposed surfaces E2a, E2b, and E2c) including the plurality of protruding portions U is, for example, preferably about 0.5 μm or more. With such a configuration, the adhesion between the metal terminal and the exterior material is further increased.

[0220] Further, the skewness Ssk of the exposed surface including the plurality of protruding portions U is preferably a positive value. With such a configuration, the adhesion between the metal terminal and the exterior material is further increased.

[0221] The RSm in the roughness curve in the second direction orthogonal or substantially orthogonal to the first direction may be, for example, about twice or more the RSm in the roughness curve in the first direction (scanning direction of the laser). The RSm in the roughness curve in the first direction (scanning direction of the laser) may be, for example, about 3 μm or more and about 10 μm or less.

Roughness Parameter Measurement Method

[0222] An example of a method of measuring the surface roughness parameter of the surface of the metal terminal is as follows. First, the exterior material is dissolved in a solvent to expose the surface of the metal terminal. In a case where the exterior material is an epoxy resin, for example,

a hydrocarbon-based release agent is used as the solvent. The surface roughness parameter of the exposed surface of the exposed metal terminal surface is measured using laser microscopy.

[0223] The roughness parameter relating to the line roughness such as RSm is measured by a measurement method according to JIS B0601-2001 (2013). The roughness parameters relating to surface roughness such as Sa and Ssk are measured by a measurement method based on ISO 25178. The arithmetic mean height Sa or the like at the time of surface roughness measurement is obtained by expanding the arithmetic mean height Ra (arithmetic mean height of a line) or the like on the surface. The average value of the roughness parameters measured at five locations is used as the measured value of the roughness parameter in the present example embodiment.

[0224] Laser microscopy: shape analysis laser microscope VK-X100 (available from Keyence Corporation)

[0225] Pitch: about 0.08 μm

[0226] Viewing range: about 150 μm × about 150 μm

[0227] Reference length at the time of line roughness measurement: standard Reference Length in VK-X

[0228] Cut-off: no

[0229] The exposed surface may be formed by forming a plating film on the terminal body, and then removing the plating film. The removal process may be various kinds of removal processes such as, for example, a mechanical removal process by grinding, polishing, or the like, a removal process by laser trimming, and a removal process by a plating stripping agent such as sodium hydroxide. Before the plating film is formed, a portion to be an exposed surface may be covered with a resist. In this case, the exposed surface is formed by removing the resist after the plating film is formed.

[0230] When the dimension in the length direction of the multilayer ceramic capacitor 1 including the exterior material 3 and the metal terminal 100 is defined as L, the L dimension is, for example, preferably about 3.2 mm or more and about 20 mm or less. When the dimension in the lamination direction of the multilayer ceramic capacitor 1 is defined as T, the T dimension is, for example, preferably about 1.0 mm or more and about 10 mm or less. When the dimension in the width direction of the multilayer ceramic capacitor 1 is defined as W, the W dimension is, for example, preferably about 1.5 mm or more and about 20 mm or less.

[0231] Next, an example of a method of manufacturing the multilayer ceramic capacitor 1 of the present example embodiment will be described. First, a method of manufacturing the multilayer ceramic capacitor main body 2 will be described.

[0232] A dielectric sheet for manufacturing the dielectric layer 20 and an electrically conductive paste for manufacturing the internal electrode layer 30 are provided. The electrically conductive paste for manufacturing the internal electrode and the dielectric sheet includes a binder and a solvent. Known binders and solvents may be used.

[0233] The electrically conductive paste for manufacturing the internal electrode layer 30 is printed on the dielectric sheet in a predetermined pattern by, for example, screen printing or gravure printing. Thus, the dielectric sheet in which the pattern of the first internal electrode layer 31 is formed, and the dielectric sheet in which the pattern of the second internal electrode layer 32 is formed are provided.

[0234] A predetermined number of dielectric sheets in which the pattern of the internal electrode layer is not printed are laminated (stacked), such that a portion defining and functioning as the first main surface-side outer layer portion 12 close to the first main surface TS1 is formed. The dielectric sheet in which the pattern of the first internal electrode layer 31 is printed and the dielectric sheet in which the pattern of the second internal electrode layer 32 is printed are sequentially laminated thereon, such that a portion defining and functioning as the inner layer portion 11 is formed. A predetermined number of the dielectric sheets in which the pattern of the internal electrode layer is not printed are laminated on the portion defining and functioning as the inner layer portion 11, such that a portion functioning as the second main surface-side outer layer portion 13 close to the second main surface TS2 is formed. Thus, a multilayer sheet is manufactured.

[0235] The multilayer sheets are pressed in the lamination direction by hydrostatic pressing, for example, such that a multilayer block is manufactured.

[0236] The multilayer block is cut to a predetermined size, such that a multilayer chip is cut out. At this time, corner portions and ridge portions of the multilayer chip may be rounded by, for example, barrel polishing or the like.

[0237] The multilayer chip is fired to manufacture the multilayer body 10. The firing temperature depends on the materials of the dielectric layer 20 and the internal electrode layer 30; however, the firing temperature is, for example, preferably about 900° C. or more and about 1400° C. or less.

[0238] The electrically conductive paste defining and functioning as the first base electrode layer 50A and the second base electrode layer 50B is applied to both end surfaces of the multilayer body 10. In the present example embodiment, the first base electrode layer 50A and the second base electrode layer 50B are fired layers. For example, an electrically conductive paste including a glass component and metal is applied to the multilayer body 10 by, for example, a method such as dipping. Thereafter, a firing process is performed to form the first base electrode layer 50A and the second base electrode layer 50B. The temperature of the firing process at this time is, for example, preferably about 700° C. or higher and about 900° C. or lower.

[0239] In addition, in a case in which the multilayer chip before firing and the electrically conductive paste applied to the multilayer chip are fired simultaneously, it is preferable that the fired layer is formed by firing a ceramic material added instead of a glass component. At this time, as the ceramic material to be added, it is particularly preferable to use the same type of ceramic material as the dielectric layer 20. In this case, the electrically conductive paste is applied to the multilayer chip before firing, and the multilayer chip and the electrically conductive paste applied to the multilayer chip are fired simultaneously to form the multilayer body 10 including the fired layer formed therein.

[0240] In addition, when a thin film layer is formed as the first base electrode layer 50A and the second base electrode layer 50B, a thin film layer may be formed on a portion of the first main surface TS1 and a portion of the second main surface TS2 of the multilayer body 10. The thin film layer may be, for example, a sputtered electrode fabricated by a sputtering method. In a case in which the sputtered electrode is formed on a portion of the first main surface TS1 and a portion of the second main surface TS2 of the multilayer

body 10 as the first base electrode layer 50A and the second base electrode layer 50B, a fired layer is formed on the first end surface LS1 and on the second end surface LS2. Alternatively, a plated layer, which will be described later, may be formed directly on the multilayer body 10 without forming the base electrode layer on the first end surface LS1 and the second end surface LS2.

[0241] Thereafter, the first plated layer 60A is formed on the first base electrode layer 50A. Furthermore, the second plated layer 60B is formed on the second base electrode layer 50B. In the present example embodiment, for example, the Ni plated layer and the Sn plated layer are formed as the plated layers. The Ni plated layer and the Sn plated layer are sequentially formed, for example, by a barrel plating method.

[0242] Through such a manufacturing process, the multilayer ceramic capacitor main body 2 is manufactured.

[0243] Next, an example of a method of manufacturing the first metal terminal 100A and the second metal terminal 100B will be described with reference to FIGS. 14A and 14B. FIG. 14A is a front view of the metal terminal before being folded. FIG. 14B is a view showing an opposite surface of the metal terminal before being folded.

[0244] A plating film is applied to the terminal main bodies defining the first metal terminal 100A and the second metal terminal 100B. After a plating film is formed on the surface of the base material of the terminal main body, the base material is cut along the shape of the metal terminal by shearing using a punching die or the like, for example. Thus, an exposed surface from which the surface of the base material of the terminal main body is exposed is formed on the lateral surface of the metal terminal main body. As shown in FIG. 14A, for a desired region of the surface of the metal terminal (the first front surface FS1 and the second front surface FS2), the film defining at least the outermost surface of the plating film is removed, and an exposed surface where the surface with low solder wettability is exposed is formed. Further, as shown in FIG. 14B, for a desired region of the back surface (the first opposite surface BS1 and the second opposite surface BS2) of the metal terminal, the film defining at least the outermost surface of the plating film is removed, and an exposed surface where the surface with low solder wettability is exposed is formed.

[0245] More specifically, the removal processing may be, for example, removal processing by laser trimming using a pulse laser. The plating film is removed by the removal process, and the exposed surface can be used as a base material of the metal terminal.

[0246] In addition, the outermost surface plating film may be removed by the removal process so that the base plating film remains. For example, in a case of laser trimming, it is possible to selectively remove only the Sn plating film defining and functioning as the outermost surface plating film among the Ni plating film defining and functioning as the base plating film and the Sn plating film defining and functioning as the outermost surface plating film, by adjusting conditions such as laser output.

[0247] Further, by adjusting the irradiation angle of the laser and adjusting the conditions such as the laser output, it is possible to form an exposed surface (the first exposed surfaces E1a, E1b, and E1c, and the second exposed surfaces E2a, E2b, and E2c) on which a plurality of protruding portions having the undercut shape of the present example embodiment are formed.

[0248] In addition, in order to configure at least a portion of the plurality of protruding portions U by a plurality of protruding portions U regularly arranged, the laser irradiation condition may be appropriately determined. For example, laser irradiation may be performed so that the spot interval of the laser beam is about 3 μm or more and about 10 μm or less. In this case, the plurality of continuously arranged protruding portions U are regularly arranged at intervals of about 3 μm or more and about 10 μm or less.

[0249] Next, an example of a process of bonding the multilayer ceramic capacitor main body 2 to the first metal terminal 100A and the second metal terminal 100B will be described.

[0250] The first external electrode 40A and the first metal terminal 100A are bonded to each other by the first bonding material 5A. The second external electrode 40B and the second metal terminal 100B are bonded by the second bonding material 5B. In the present example embodiment, the first bonding material 5A and the second bonding material 5B are solder, for example. For example, when the bonding is performed by reflow soldering, the first bonding material 5A and the second bonding material 5B are heated, for example, at a temperature of about 270° C. or more and about 290° C. or less for about 30 seconds or more.

[0251] The heating during the reflow process melts the first bonding material 5A and the second bonding material 5B. At this time, since the exposed surface E1b is provided on the surface of the first rising portion 120A of the first metal terminal 100A, which is a surface opposed to the first surface S1 of the multilayer ceramic electronic component main body 2, the first bonding material 5A is difficult to spread along the first rising portion 120A of the first metal terminal 100A. In addition, since the exposed surface E2b is provided on the surface of the second rising portion 120B of the second metal terminal 100B, which is a surface opposed to the second surface S2 of the multilayer ceramic electronic component main body 2, the second bonding material 5B is difficult to spread along the rising portion 120B of the second metal terminal 100B. Similarly, the exposed surface E1a and the exposed surface E2a have a function of preventing solder from spreading during reflow. Similarly, the exposed surface E1c and the exposed surface E2c have a function of preventing solder from spreading during reflow. Similarly, the surface of the first base material 100Aa located on the first terminal lateral surface TSS1 and the surface of the second base material 100Ba located on the second terminal lateral surface TSS2 have a function of preventing solder from spreading during reflow.

[0252] After the heating, the first bonding material 5A is solidified in a state where the gap portion G remains between the first rising portion 120A of the first metal terminal 100A and the first surface S1 of the multilayer ceramic capacitor main body 2 on the first end surface LS1 such that the multilayer ceramic capacitor main body 2 and the first metal terminal 100A are bonded to each other. In addition, the second bonding material 5B is solidified in a state where the gap portion G remains between the second rising portion 120B of the second metal terminal 100B and the second surface S2 on the second end surface LS2 of the multilayer ceramic capacitor main body 2 such that the multilayer ceramic capacitor main body 2 and the second metal terminal 100B are bonded to each other. With such a configuration, it is possible to fill the gap portion G more reliably with the exterior material 3 in the subsequent step.

[0253] Next, an example of a process of covering the multilayer ceramic capacitor main body 2, the first bonding material 5A and the second bonding material 5B, a portion of the first metal terminal 100A, and a portion of the second metal terminal 100B with the exterior material 3 will be described.

[0254] The exterior material 3 is formed by, for example, a transfer molding method. Specifically, the multilayer ceramic capacitor before being covered with the exterior material 3, that is, the multilayer ceramic capacitor main body 2 to which the metal terminal 100 is bonded via the bonding material 5, is arranged in a mold, and then the resin of the exterior material 3 is filled in the mold, and the resin is cured. Thus, the exterior material 3 is provided so as to cover the multilayer ceramic capacitor main body 2, the first bonding material 5A and the second bonding material 5B, a portion of the first metal terminal 100A, and a portion of the second metal terminal 100B. At this time, the gap portion G can also be filled with the exterior material 3.

[0255] Finally, if there is an unnecessary portion in the metal terminal 100, the unnecessary portion is cut using, for example, a stamping die or the like. Then, the metal terminal 100 is bent into a desired shape using, for example, a bending die or the like. Thus, the metal terminal 100 may be formed by bending. That is, each connection portion of the metal terminal 100 formed by bending may be formed by bending. The bending process is partially performed before molding the exterior material 3.

[0256] According to the above-described manufacturing method, the multilayer ceramic capacitor 1 according to an example embodiment of the present invention is manufactured.

[0257] FIGS. 15A and 15B each show a mounting structure 300 of the multilayer ceramic capacitor 1. FIG. 15A is an external perspective view of a mounting structure 300 in which the multilayer ceramic capacitor 1 of the present example embodiment is mounted on a mounting substrate 310. FIG. 15B is a view corresponding to FIG. 6, and is a virtual arrow view when the mounting structure 300 of the multilayer ceramic capacitor 1 of FIG. 15A is viewed in the direction of the arrow XIIIIB.

[0258] Thereafter, the multilayer ceramic capacitor 1 which is covered with the exterior material 3 and completed is reflow-mounted as a component on the mounting substrate 310 via a substrate mounting bonding material 320.

[0259] More specifically, the first metal terminal 100A is bonded to a wiring member 312 provided on the mounting surface 311 of the mounting substrate 310 via the substrate mounting bonding material 320. The second metal terminal 100B is bonded to the wiring member 312 provided on the mounting surface 311 of the mounting substrate 310 via the substrate mounting bonding material 320.

[0260] At this time, the bonding material 5 may melt and the volume of the bonding material 5 may expand. However, with the configuration including the plurality of exposed surfaces shown in the present example embodiment, it is possible to reduce or prevent the occurrence of solder splash or other problems.

[0261] Hereinafter, a modified example embodiment of the multilayer ceramic capacitor 1 of the present example embodiment will be described. In the following description, the same components as those in the above example embodiments are denoted by the same reference numerals, and detailed descriptions thereof will be omitted. FIG. 16A is a

view showing a modified example of the multilayer ceramic capacitor **1** of the present example embodiment, and corresponds to FIG. 2. FIG. 16B is an arrow view when the multilayer ceramic capacitor **1** of FIG. 16A is viewed in the direction of the arrow XVB, and corresponds to FIG. 4.

[0262] In the present modified example, the configuration of the metal terminal is different from that of the above-described example embodiments. The metal terminal of the modified example includes a first metal terminal **200A** and a second metal terminal **200B**.

[0263] The configuration of a portion of the first metal terminal **200A** provided inside the exterior material **3** is the same or substantially the same as the configuration of the first metal terminal **100A** of the above example embodiment. The configuration of the portion of the second metal terminal **200B** provided inside the exterior material **3** is the same or substantially the same as the configuration of the second metal terminal **100B** of the above example embodiment.

[0264] The first metal terminal **200A** includes a first extension portion **230A**, a first falling portion **240A**, and a first mounting portion **250A**. The first extension portion **230A** is connected to the first falling portion **240A** immediately after protruding from the first end surface MLS1 of the exterior material **3**. The connection portion between the first extension portion **230A** and the first falling portion **240A** is formed by bending at a right angle or substantially a right angle. The first falling portion **240A** extends in a direction orthogonal or substantially orthogonal to the mounting surface toward the mounting surface. The first mounting portion **250A** extends along the mounting surface toward the middle side in the length direction L of the multilayer ceramic capacitor **1**.

[0265] The second metal terminal **200B** includes a second extension portion **230B**, a second falling portion **240B**, and a second mounting portion **250B**. The second extension portion **230B** is connected to the second falling portion **240B** immediately after protruding from the second end surface MLS2 of the exterior material **3**.

[0266] The connection portion between the second extension portion **230B** and the second falling portion **240B** is formed by bending at a right angle or substantially a right angle. The second falling portion **240B** extends in a direction orthogonal or substantially orthogonal to the mounting surface toward the mounting surface. The second mounting portion **250B** extends along the mounting surface toward the middle side in the length direction L of the multilayer ceramic capacitor **1**.

[0267] This makes it possible to shorten the dimension L8 in the length direction of the multilayer ceramic capacitor **1** including the first metal terminal **200A** and the second metal terminal **200B**. Therefore, it is possible to reduce the mounting area required for mounting the multilayer ceramic capacitor **1** on the mounting substrate.

[0268] Also in this case, the separation distance L7 between the end of the first mounting portion **250A** of the first metal terminal **200A** and the end of the second mounting portion **250B** of the second metal terminal **200B** is preferably longer than the separation distance L3 between the first external electrode **40A** and the second external electrode **40B** of the multilayer ceramic capacitor main body **2** shown in FIG. 7.

[0269] The first mounting portion **250A** may extend in parallel or substantially in parallel to the mounting surface along the mounting surface, or may extend to be sloped

away from the mounting surface toward the middle in the length direction L of the multilayer ceramic capacitor **1**. The second mounting portion **250B** may extend in parallel or substantially in parallel to the mounting surface along the mounting surface, or may extend to be sloped away from the mounting surface toward the middle in the length direction L of the multilayer ceramic capacitor **1**. With such a configuration, when the multilayer ceramic capacitor **1** is mounted on the mounting substrate, it is possible to extend the bonding material to this portion, and it is possible to increase the strength in mounting. In addition, it is possible to stably provide the multilayer ceramic capacitor **1** on the mounting surface of the mounting substrate. The slope angle θ is, for example, preferably about 1° or more and about 40° or less.

[0270] The metal terminal may include an exposed surface at a position different from that of the first example embodiment. FIG. 16A shows an example of the arrangement position of an additional exposed surface ES3 of the first metal terminal **200A** and the arrangement position of an additional exposed surface ES4 of the second metal terminal **200B**.

[0271] The additional exposed surface ES3 is provided on a surface of the first falling portion **240A** of the first metal terminal **200A**, which is a surface opposed to the first sloped surface MLS1A of the exterior material **3** of the multilayer ceramic electronic component **1**. The additional exposed surface ES3 may also be provided on a surface of the first mounting portion **250A** opposite to the mounting surface, that is, a surface opposed to the first main surface MTS1 of the exterior material **3**.

[0272] The additional exposed surface ES4 is provided on a surface of the second falling portion **240B** of the second metal terminal **200B**, which is a surface opposed to the second sloped surface MLS2A of the exterior material **3** of the multilayer ceramic capacitor **1**. The additional exposed surface ES4 may also be provided on a surface of the second mounting portion **250B** opposite to the mounting surface, that is, a surface opposed to the first main surface MTS1 of the exterior material **3**.

[0273] With such a configuration, when the multilayer ceramic capacitor **1** is mounted, it is possible to prevent the solder from entering between the first metal terminal **200A** and the exterior material **3** and between the second metal terminal **200B** and the exterior material **3**.

[0274] In the multilayer ceramic capacitor main body **2** of the present example embodiment, the plurality of the first internal electrode layers **31** and the plurality of the second internal electrode layers **32** are provided alternately in the height direction T of the multilayer body **10**. However, the configuration of the multilayer ceramic capacitor main body **2** is not limited thereto. The plurality of the first internal electrode layers **31** and the plurality of the second internal electrode layers **32** may be alternately provided in the width direction W of the multilayer body **10**.

[0275] In this case, the first extension portion of each of the first internal electrode layers **31** may extend out toward the first main surface TS1 adjacent to the first end surface LS1, and the first external electrode **40A** may be provided only on the first main surface TS1 adjacent to the first end surface LS1. That is, the first end surface LS1 may not be provided with the first external electrode **40A**. In such a case, the first surface S1 on the first end surface LS1 of the multilayer ceramic capacitor main body **2** includes the first

end surface LS1 of the multilayer body 10. Further, the second extension portion of the second internal electrode layers 32 may extend out toward the first main surface TS1 adjacent to the second end surface LS2, and the second external electrode 40B may be provided only on the first main surface TS1 adjacent to the second end surface LS2. That is, the second end surface LS2 may not be provided with the second external electrode 40B. In such a case, the first surface S1 on the second end surface LS2 of the multilayer ceramic capacitor main body 2 includes the second end surface LS2 of the multilayer body 10. In this case, the bonding material 5 is difficult to spread in the gap portion G.

[0276] In the present example embodiment, one multilayer ceramic capacitor main body 2 is covered with the exterior material 3 to provide the multilayer ceramic capacitor 1. However, the present invention is not limited thereto. The multilayer ceramic capacitor main body 2 defining and functioning as the plurality of the multilayer ceramic electronic component main bodies may be covered with the exterior material 3 to provide the multilayer ceramic capacitor 1 defining and functioning as a multilayer ceramic electronic component. For example, a plurality of the multilayer ceramic capacitor main body 2 provided in parallel or substantially in parallel may be covered with the exterior material 3 to provide the multilayer ceramic capacitor 1. For example, multilayer ceramic capacitor main body 2 stacked in two or more stages may be covered with the exterior material 3 to provide the multilayer ceramic capacitor 1.

[0277] The configuration of the multilayer ceramic capacitor main body is not limited to the configuration shown in FIGS. 7 to 10. For example, the multilayer ceramic capacitor main body may be a multilayer ceramic capacitor with a two-portion structure, a three-portion structure, or a four-portion structure as shown in FIGS. 17A, 17B, and 17C.

[0278] The multilayer ceramic capacitor main body 2 shown in FIG. 17A is a multilayer ceramic capacitor main body 2 with a two-portion structure, and includes, as the internal electrode layer 30, in addition to the first internal electrode layer 33 and the second internal electrode layer 34, a floating internal electrode layer 35 that is not exposed at either of the first end surface LS1 and the second end surface LS2. The multilayer ceramic capacitor main body 2 shown in FIG. 17B is a multilayer ceramic capacitor main body 2 with a three-portion structure including a first floating internal electrode layer 35A and a second floating internal electrode layer 35B as the floating internal electrode layers 35. The multilayer ceramic capacitor main body 2 shown in FIG. 17C is a multilayer ceramic capacitor main body 2 with a four-portion structure including a first floating internal electrode layer 35A, a second floating internal electrode layer 35B, and a third floating internal electrode layer 35C as the floating internal electrode layers 35. As described above, by providing the floating internal electrode layer 35 defining and functioning as the internal electrode layer 30, the multilayer ceramic capacitor main body 2 has a structure in which the counter electrode portion is divided into a plurality of portions. With such a configuration, a plurality of capacitor components are provided between the opposing internal electrode layers 30, and these capacitor components are connected in series. Therefore, the voltage applied to each capacitor component becomes low, and the breakdown voltage of the multilayer ceramic capacitor main body 2 can be increased. The multilayer ceramic capacitor main body 2

of the present example embodiment may have a multiple-portion structure of four or more portions.

[0279] In addition, the multilayer ceramic capacitor main body 2 may be of a two-terminal capacitor including two external electrodes, or may be of a multi-terminal capacitor including a large number of external electrodes.

[0280] According to the multilayer ceramic capacitor 1 of the present example embodiment, the following advantageous effects are obtained.

[0281] (1) A multilayer ceramic capacitor 1 according to an example embodiment of the present invention includes the multilayer ceramic capacitor main body 2 (the multilayer ceramic electronic component main body 2) including the multilayer body 10 including the plurality of dielectric layers 20 (the ceramic layers 20) and the plurality of internal electrode layers 30 (the internal conductive layers 30) that are each laminated on a corresponding one of the plurality of dielectric layers 20, the first main surface TS1 and the second main surface TS2 opposed to each other in the height direction T, the first lateral surface WS1 and the second lateral surface WS2 opposed to each other in the width direction W orthogonal or substantially orthogonal to the height direction T, and the first end surface LS1 and the second end surface LS2 opposed to each other in the length direction L orthogonal or substantially orthogonal to the height direction T and the width direction W, the first external electrode 40A on the first end surface LS1, and the second external electrode 40B on the second end surface LS2, the first metal terminal 100A connected to the first external electrode 40A via the bonding material 5 (the first bonding material 5A), the second metal terminal 100B connected to the second external electrode 40B via the bonding material 5 (the second bonding material 5B), and the exterior material 3 covering the multilayer ceramic capacitor main body 2, the bonding material 5, a portion of the first metal terminal 100A, and a portion of the second metal terminal 100B. The first metal terminal 100A includes the first bonding surface 110A1 bonded to the bonding material 5 (the first bonding material 5A), and the first contact surface CS1 in contact with the exterior material 3. The first contact surface CS1 includes a first outermost surface plating film 100Ab2 (a first outermost surface metal film 100Ab2) on at least a portion of a surface of the first contact surface CS1. The second metal terminal 100B includes a second bonding surface 110B1 bonded to the bonding material 5 (the second bonding material 5B), and a second contact surface CS2 in contact with the exterior material 3. The second contact surface CS2 includes a second outermost surface plating film 100Bb2 (a second outermost surface metal film 100Bb2) on at least a portion of a surface of the second contact surface CS2. The first contact surface CS1 includes the plurality of first exposed surfaces defining and functioning as the exposed surfaces E1a, E1b, and E1c, each including an exposed surface of a metal different from the first outermost surface plating film 100Ab2. The second contact surface CS2 includes the plurality of second exposed surfaces defining and functioning as the exposed surfaces E2a, E2b, and E2c, each including an exposed surface of a metal different from the second outermost surface plating film 100Bb2. The plurality of first exposed surfaces E1a, E1b, and E1c and the plurality of second exposed surfaces E2a, E2b, and E2c each include a plurality of protruding portions U, and the plurality of protruding portions U each have an undercut shape. With

such a configuration, it is possible to provide multilayer ceramic electronic components that are each able to reduce or prevent peeling at an interface between a metal terminal and an exterior material and reduce or prevent a decrease in reliability of the electronic component.

[0282] (2) In a multilayer ceramic capacitor **1** according to an example embodiment of the present invention, the plurality of protruding portions **U** are continuously provided in-plane with the plurality of first exposed surfaces (**E1a**, **E1b**, and **E1c**) and the plurality of second exposed surfaces (**E2a**, **E2b**, and **E2c**). With such a configuration, an anchor effect is generated between the exterior material and the plurality of protruding portions provided on the exposed surfaces of the metal terminal, such that it is possible to increase the adhesion between the metal terminal and the exterior material.

[0283] (3) In a multilayer ceramic capacitor **1** according to an example embodiment of the present invention, a skewness **Ssk** of a surface of each of the plurality of first exposed surfaces (**E1a**, **E1b**, and **E1c**) or a surface of each of the plurality of second exposed surfaces (**E2a**, **E2b**, and **E2c**) including the plurality of protruding portions **U** is a positive value. This further increases the adhesion between the metal terminal and the exterior material.

[0284] (4) In a multilayer ceramic capacitor **1** according to an example embodiment of the present invention, the first metal terminal **100A** includes the first base material **100Aa** and the first plating film **100Ab** on a surface of the first base material **100Aa**. The first outermost surface plating film **100Ab2** includes at least an outermost surface portion of the first plating film **100Ab**. The second metal terminal **100B** includes the second base material **100Ba** and the second plating film **100Bb** on a surface of the second base material **100Ba**. The second outermost surface plating film **100Bb2** includes at least an outermost surface portion of the second plating film **100Bb**. With such a configuration, it is possible to provide a multilayer ceramic electronic component that is able to appropriately reduce or prevent excessive outflow of solder and reduce or prevent the occurrence of solder splash, while maintaining wettability of solder in a necessary portion.

[0285] (5) In a multilayer ceramic capacitor **1** according to an example embodiment of the present invention, the first plating film **100Ab** includes the first base plating film **100Ab1** that covers a surface of the first base material **100Aa** and the first outermost surface plating film **100Ab2** that covers a surface of the first base plating film **100Ab1**. The first outermost surface metal film is the first outermost surface plating film **100Ab2**. The second plating film **100Bb** includes the second base plating film **100Bb1** that covers a surface of the second base material **100Ba** and the second outermost surface plating film **100Bb2** that covers a surface of the second base plating film **100Bb1**. The second outermost surface metal film is the second outermost surface plating film **100Bb2**. With such a configuration, it is possible to provide a multilayer ceramic electronic component that is able to appropriately reduce or prevent excessive outflow of solder and reduce or prevent the occurrence of solder splash, while maintaining wettability of solder in a necessary portion.

[0286] (6) In a multilayer ceramic capacitor according to an example embodiment of the present invention, the first outermost surface plating film **100Ab2** is a Sn plating film, and the second outermost surface plating film **100Bb2** is a

Sn plating film. As a result, the solder wettability of the necessary portion can be appropriately ensured.

[0287] (7) In a multilayer ceramic capacitor **1** according to an example embodiment of the present invention, the first base plating film **100Ab1** is a Ni plating film, and the second base plating film **100Bb1** is a Ni plating film. With such a configuration, it is possible to provide a multilayer ceramic electronic component that is able to appropriately reduce or prevent the outflow of solder and reduce or prevent the occurrence of solder splash, while maintaining solder barrier performance.

[0288] (8) In a multilayer ceramic capacitor **1** according to an example embodiment of the present invention, each of the first exposed surfaces **E1a**, **E1b**, and **E1c** defining and functioning as the first exposed surface is the first base plating film **100Ab1** of the first metal terminal **100A**, and each of the second exposed surfaces **E2a**, **E2b**, and **E2c** defining and functioning as the second exposed surface is the second base plating film **100Bb1** of the second metal terminal **100B**. With such a configuration, since it is possible to lower the solder wettability of the exposed portion, it is possible to provide a multilayer ceramic electronic component that is able to appropriately reduce or prevent the outflow of the bonding material and reduce or prevent the occurrence of solder splash.

[0289] (9) In a multilayer ceramic capacitor according to an example embodiment of the present invention, each of the first exposed surfaces **E1a**, **E1b**, and **E1c** is the first base material **100Aa** of the first metal terminal **100A**, and each of the second exposed surfaces **E2a**, **E2b**, and **E2c** is the second base material **100Ba** of the second metal terminal **100B**. With such a configuration, since it is possible to further lower the solder wettability of the exposed portion, it is possible to provide a multilayer ceramic electronic component that is able to appropriately reduce or prevent the outflow of solder and reduce or prevent the occurrence of solder splash.

[0290] (10) In a multilayer ceramic capacitor **1** according to an example embodiment of the present invention, each of the first metal terminal **100A** and the second metal terminal **100B** is a metal terminal mounted on a mounting surface **311** of a mounting substrate **310** on which the multilayer ceramic capacitor **1** is mounted. The first main surface **TS1** of the multilayer body **10** is a surface opposed to the mounting surface **311**. The first external electrode **40A** is provided at least on a portion of the first main surface **TS1** adjacent to the first end surface **LS1**. The second external electrode **40B** is provided at least on a portion of the first main surface **TS1** adjacent to the second end surface **LS2**. The first metal terminal **100A** includes the first bonding portion **110A** that is opposed to the first main surface **TS1** and connected to the first external electrode **40A**, the first rising portion **120A** that is connected to the first bonding portion **110A** and extends away from the mounting surface **311**, and the first extension portion **130A** that is connected to the first rising portion **120A** and extends away from the multilayer ceramic capacitor main body **2**. The second metal terminal **100B** includes the second bonding portion **110B** that is opposed to the first main surface **TS1** and connected to the second external electrode **40B**, the second rising portion **120B** that is connected to the second bonding portion **110B** and extends away from the mounting surface **311**, and the second extension portion **130B** that is connected to the second rising portion **120B** and extends away from the multilayer ceramic

capacitor main body 2. With such a configuration, it is possible to lengthen the path of the solder spreading, and it is possible to reliably prevent the solder from flowing out to an unnecessary portion.

[0291] (11) In a multilayer ceramic capacitor 1 according to an example embodiment of the present invention, the first metal terminal 100A is plate shaped and includes the first front surface FS1 adjacent to the first bonding surface 110A1 to which the first external electrode 40A is bonded, the first opposite surface BS1 which is a surface opposite to the first front surface FS1, and the first terminal lateral surface TSS1 connecting the first front surface FS1 and the first opposite surface BS1. The second metal terminal 100B is plate shaped and includes the second front surface FS2 adjacent to the second bonding surface 110B1 to which the second external electrode 40B is bonded, the second opposite surface BS2 which is a surface opposite to the second front surface FS2, and the second terminal lateral surface TSS2 connecting the second front surface FS2 and the second opposite surface BS2. This makes it possible to easily form the exposed surface. For example, the exposed surface can be easily formed by processing using laser trimming, a punching die, or the like.

[0292] (12) In a multilayer ceramic capacitor 1 according to an example embodiment of the present invention, the exposed surfaces E1a, E1b, and E1c defining and functioning as the first exposed surfaces provided on the first contact surface CS1 are respectively separated in the width direction by a hole or a notch provided in the first metal terminal 100A. The second exposed surfaces E2a, E2b, and E2c provided on the second contact surface CS2 are respectively separated in the width direction by a hole or a notch provided in the second metal terminal 100B. With such a configuration, in a case where the exposed surface is formed by processing such as laser trimming, for example, it is possible to effectively prevent the solder from flowing out, while reducing the amount of processing for processing the exposed surface.

[0293] In addition, in the example embodiments described above, a multilayer ceramic capacitor including a dielectric ceramic has been exemplified as the multilayer ceramic electronic components. However, multilayer ceramic electronic components according to example embodiments of the present invention are not limited thereto, and are applicable to various multilayer ceramic electronic components such as, for example, a piezoelectric component including a piezoelectric ceramic, a thermistor including a semiconductor ceramic, and an inductor including a magnetic ceramic. Examples of the piezoelectric ceramics include PZT (lead zirconate titanate) ceramics, examples of semiconductor ceramics include spinel ceramics, and examples of magnetic ceramics include ferrite.

[0294] The present invention is not limited to the configurations of the example embodiments described above, and can be appropriately modified and applied without departing from the scope of the present invention. The present invention also includes combinations of two or more of the individual desirable configurations described in the example embodiments.

[0295] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the

present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

What is claimed is:

1. A multilayer ceramic electronic component comprising:

a multilayer ceramic electronic component main body including a multilayer body including a plurality of ceramic layers, and a plurality of internal conductive layers laminated on a corresponding one of the plurality of ceramic layers, a first main surface and a second main surface opposed to each other in a height direction, a first lateral surface and a second lateral surface opposed to each other in a width direction orthogonal or substantially orthogonal to the height direction, and a first end surface and a second end surface opposed to each other in a length direction orthogonal or substantially orthogonal to the height direction and the width direction, a first external electrode on the first end surface, and a second external electrode on the second end surface;

a first metal terminal connected to the first external electrode via a bonding material;

a second metal terminal connected to the second external electrode via the bonding material; and

an exterior material covering the multilayer ceramic electronic component main body, the bonding material, a portion of the first metal terminal, and a portion of the second metal terminal; wherein

the first metal terminal includes:

a first bonding surface bonded to the bonding material; and

a first contact surface in contact with the exterior material and including a first outermost surface metal film on at least a portion of a surface of the first contact surface;

the second metal terminal includes:

a second bonding surface bonded to the bonding material; and

a second contact surface in contact with the exterior material and including a second outermost surface metal film on at least a portion of a surface of the second contact surface;

the first contact surface includes a plurality of first exposed surfaces each including an exposed surface of a metal different from the first outermost surface metal film;

the second contact surface includes a plurality of second exposed surfaces each including an exposed surface of a metal different from the second outermost surface metal film;

the plurality of first exposed surfaces and the plurality of second exposed surfaces each include a plurality of protruding portions; and

the plurality of protruding portions each have an undercut shape.

2. The multilayer ceramic electronic component according to claim 1, wherein the plurality of protruding portions are continuously provided in-plane with the plurality of first exposed surfaces and the plurality of second exposed surfaces.

3. The multilayer ceramic electronic component according to claim 1, wherein a skewness of a surface of each of the plurality of first exposed surfaces or a surface of each of the plurality of second exposed surfaces including the plurality of protruding portions is a positive value.

4. The multilayer ceramic electronic component according to claim 1, wherein

the first metal terminal includes a first base material and a first plating film on a surface of the first base material, and the first outermost surface metal film includes at least an outermost surface portion of the first plating film; and

the second metal terminal includes a second base material and a second plating film on a surface of the second base material, and the second outermost surface metal film includes at least an outermost surface portion of the second plating film.

5. The multilayer ceramic electronic component according to claim 4, wherein

the first plating film includes a first base plating film covering a surface of the first base material and a first outermost surface plating film covering a surface of the first base plating film, the first outermost surface metal film being the first outermost surface plating film; and the second plating film includes a second base plating film covering a surface of the second base material and a second outermost surface plating film covering a surface of the second base plating film, the second outermost surface metal film being the second outermost surface plating film.

6. The multilayer ceramic electronic component according to claim 5, wherein

the first outermost surface plating film includes a Sn plating film; and the second outermost surface plating film includes a Sn plating film.

7. The multilayer ceramic electronic component according to claim 5, wherein

the first base plating film includes a Ni plating film; and the second base plating film includes a Ni plating film.

8. The multilayer ceramic electronic component according to claim 5, wherein

each of the first exposed surfaces is defined by the first base plating film of the first metal terminal; and each of the second exposed surfaces is defined by the second base plating film of the second metal terminal.

9. The multilayer ceramic electronic component according to claim 4, wherein

each of the first exposed surfaces is defined by the first base material of the first metal terminal; and each of the second exposed surfaces is defined by the second base material of the second metal terminal.

10. The multilayer ceramic electronic component according to claim 1, wherein

each of the first metal terminal and the second metal terminal is mounted on a mounting surface of a mounting substrate on which the multilayer ceramic electronic component is mounted;

the first main surface of the multilayer body is opposed to the mounting surface;

the first external electrode is provided at least on a portion of the first main surface adjacent to the first end surface; the second external electrode is provided at least on a portion of the first main surface adjacent to the second end surface;

the first metal terminal includes a first bonding portion opposed to the first main surface and connected to the first external electrode, a first rising portion connected to the first bonding portion and extending away from

the mounting surface, and a first extension portion connected to the first rising portion and extending away from the multilayer ceramic electronic component; and the second metal terminal includes a second bonding portion opposed to the first main surface and connected to the second external electrode, a second rising portion connected to the second bonding portion and extending away from the mounting surface, and a second extension portion connected to the second rising portion and extending away from the multilayer ceramic electronic component.

11. The multilayer ceramic electronic component according to claim 1, wherein

the first metal terminal is plate shaped and includes a first front surface adjacent to the first bonding surface to which the first external electrode is bonded, a first opposite surface opposite to the first front surface, and a first terminal lateral surface connecting the first front surface and the first opposite surface; and

the second metal terminal is plate shaped and includes a second front surface adjacent to the second bonding surface to which the second external electrode is bonded, a second opposite surface opposite to the second front surface, and a second terminal lateral surface connecting the second front surface and the second opposite surface.

12. The multilayer ceramic electronic component according to claim 1, wherein

the first exposed surfaces on the first contact surface are respectively separated in the width direction by a hole or a notch provided in the first metal terminal; and

the second exposed surfaces on the second contact surface are respectively separated in the width direction by a hole or a notch provided in the second metal terminal.

13. The multilayer ceramic electronic component according to claim 1, wherein each of the first and second metal terminal is defined by a plate-shaped lead frame.

14. The multilayer ceramic electronic component according to claim 10, wherein

the first metal terminal further includes a first falling portion connected to the first extension portion and extending toward the mounting surface; and

the second metal terminal further includes a second falling portion connected to the second extension portion and extending toward the mounting surface.

15. The multilayer ceramic electronic component according to claim 1, wherein a minimum distance between a lateral surface of the exterior material to a surface of the multilayer ceramic capacitor main body is about 100 μm or more and about 4000 μm or less.

16. The multilayer ceramic electronic component according to claim 1, wherein a minimum distance between an end surface of the exterior material and a surface of the multilayer ceramic capacitor main body is about 300 μm or more and about 5000 μm or less.

17. The multilayer ceramic electronic component according to claim 1, wherein the exterior material includes resin.

18. The multilayer ceramic electronic component according to claim 10, wherein

the first rising portion is sloped away from the first end surface from a connection portion with the first bonding portion toward a connection portion with the first extension portion; and

the second rising portion is sloped away from the second end surface from a connection portion with the second bonding portion toward a connection portion with the second extension portion.

19. The multilayer ceramic electronic component according to claim **18**, wherein

an angle between the first rising portion and the first end surface is about 1° or more and about 40° or less; and
an angle between the second rising portion and the second end surface is about 1° or more and about 40° or less.

20. The multilayer ceramic electronic component according to claim **10**, wherein

the first bonding portion includes a first notch extending from an end of the first bonding portion to a middle of the first rising portion; and

the second bonding portion includes a second notch extending from an end of the second bonding portion to a middle of the second rising portion.

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